

EXTENDED RESEARCH PROJECT ARTIFICIAL INTELLIGENCE



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Viewpoint Diversity for UAV Object Detection

From Controlled Synthetic Training to Multiview Diagnostics and Real-World Transfer

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September 7, 2026

Acknowledgements

This thesis has been a journey that was not always as straightforward as I initially imagined. There were changes in direction, challenges, and moments where the next step was not immediately clear. I am proud of what this process has become and especially grateful for the people who supported, challenged, and guided me along the way.

First, I would like to sincerely thank Marcos De Paulo Bueno for supervising me throughout the first nine months of this research. His guidance helped me shape the initial direction of the thesis and provided an important foundation for the work that followed.

I would also like to thank Roy Lindelauf for his supervision at the DSCE Center, as well as everyone else at the DSCE Center who took the time to discuss my work, answer my questions, and help me refine the direction of my research. Beyond the technical aspects, I am especially grateful for the opportunity to experience how research is conducted within the defence domain and how research questions can emerge from practical challenges and contribute to real-world applications.

I am extremely grateful to Matan, who took over the supervision in June and guided me through the final months of the project. His fresh perspective helped me further develop the research and encouraged me to explore additional experiments that made the drone research more interesting and complete. I particularly appreciate the time he invested in critically reading my work, discussing the results with me, and providing feedback. Perhaps most importantly, his enthusiasm and approach to research. He reminded me that research can also be about exploring interesting questions, trying unexpected directions, and being genuinely curious about what the results might reveal. Even results that initially appear less exciting can still be informative. Learning to look beyond that and instead ask what it could teach me was an important part of rediscovering my appreciation for research. For that renewed perspective, I am especially grateful.

Finally, I want to thank the supervisors, friends, and family who have been alongside me throughout this process. This thesis did not follow a perfectly straight path from beginning to end, and there were certainly some bumps along the way. Alongside the research itself, this period also included personal challenges and milestones, including training for and completing a half-distance triathlon. I am especially grateful for sport as a way to maintain my energy, clear my head, and occasionally release some frustration, and for the great group of people around me who were always willing to train, struggle, and celebrate these moments together.

Completing this thesis marks the end of a challenging but rewarding period, and I am proud of what I have achieved. More than anything, I am grateful that I did not have to make the journey alone. Thank you.

Abstract

Object detection from unmanned aerial vehicles (UAVs) is sensitive to camera-object geometry: the same object can appear substantially different when observed from another direction, height, or distance. In real UAV datasets viewpoint variation typically occurs together with changes in object identity, background, illumination, scale, and occlusion, making it difficult to isolate the effect of viewpoint itself. This thesis analyzes viewpoint diversity in UAV object detection through a combination of controlled synthetic data, public UAV datasets, and self-acquired real-world UAV imagery, with the synthetic benchmark enabling viewpoint to be treated as an explicit experimental variable. Its core synthetic benchmark contains 10 object classes observed across a structured grid of eight azimuths, three elevation levels, and three camera-to-object distances. Three detectors (YOLOv8n, YOLOv8l, and Faster R-CNN) were trained under progressively broader viewpoint-exposure regimes and evaluated using a common, common, image-disjoint synthetic test split covering the full viewpoint grid, with YOLOv8l obtaining the highest test-set point estimate and therefore being selected for the subsequent analyses.

The full-viewpoint M4 training regime achieved the highest test-set performance within all three detector families, while equal-budget controls supported a benefit of distributing a fixed image budget across a broader range of viewpoints. Fixed-detector analysis shows that viewpoint sensitivity remains structured even after broad viewpoint training. Within the controlled, target-centered capture regime, elevation produced the largest and most directionally consistent cross-class effect, whereas azimuth showed weaker and more class-dependent variation and radius had a smaller, largely scale-related influence. Inference-time multiview experiments showed that the largest marginal gain in detection came from moving between one and two available observations, with further, comparable gains from accumulation-based fusion, especially for weak, small, or poorly localized detections. Shapley-based attribution was used to quantify these view-wise contributions.

Finally, synthetic-to-real experiments showed that the strongest synthetic detector transferred poorly to real UAV imagery without adaptation, whereas fine-tuning the synthetic-trained detector on only 80 annotated real UAV training images substantially improved performance on a separate held-out real test set. Importantly, domain adaptation and multiview availability played different roles: real-image fine-tuning accounted for the major improvement in basic real-world detection performance, while access to additional viewpoints provided a smaller further increase in the probability of obtaining a high-quality target observation. These experiments demonstrate that a synthetic-trained detector can be substantially adapted to the real domain. However, because no real-only training baseline was included, they do not isolate how much of the final performance is attributable specifically to synthetic pretraining rather than to the subsequent real-image supervision.

Overall, this thesis provides a controlled framework for disentangling three distinct roles of viewpoint in UAV object detection: (i) the geometric coverage required during training, (ii) the observation-dependent sensitivity of a trained detector, and (iii) the informational value of additional observations at inference time. In doing so, it distinguishes the benefits of training-time viewpoint diversity, inference-time viewpoint selection, multiview observation, and real-domain adaptation, while establishing which controlled synthetic findings remain relevant when moving toward real UAV imagery.

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Chapter 1

Introduction

Object detection from unmanned aerial vehicles is strongly shaped by camera-object geometry. UAVs are able to observe the same target from different azimuths, elevations, and distances. This flexibility is operationally useful, but can make visual evidence unstable across views. As the camera moves, objects in the point of view change in scale, aspect ratio, silhouette, visible surfaces, texture. Some viewpoints expose discriminative cues, while others hide or compress them.

Object detection, the task of identifying, localizing, and classifying objects in images, is one of the central problems in computer vision (Zaidi et al., 2022). Object detection typically forms the first stage of a broader UAV perception pipeline that includes tracking, counting, sensor fusion, and downstream decision-making (Tang et al., 2024; Krump and Stütz, 2023). Reliable UAV detection underpins applications such as surveillance (Ramachandran and Sangaiah, 2021), infrastructure inspection (Lyu et al., 2025), traffic monitoring (Khanpour et al., 2025), precision agriculture (Honkavaara et al., 2013), environmental observation (Tang et al., 2024), and disaster response (Erdelj et al., 2017). Object detection can be complicated by small, weakly textured, and partially occluded targets embedded in cluttered scenes (Zhu et al., 2021a). This can be further complicated by viewpoint variation: the same object may appear frontal, lateral, oblique, or near top-down depending on camera trajectory, with measurable effects on detection performance (Wu et al., 2019; Jin et al., 2025).

Public UAV benchmarks provide realistic aerial variation, but viewpoint typically changes together with object identity, scale, background, occlusion, illumination, and other acquisition conditions (Du et al., 2018; Zhu et al., 2021a; Bozcan and Kayacan, 2020). This makes it difficult to isolate the independent effect of camera-object geometry. More recent viewpoint-aware benchmarks confirm that camera pose and perspective matter, and synthetic environments offer stronger experimental control (Lee et al., 2022). However, existing approaches generally provide partial viewpoint control rather than dense, repeated, object-centric sampling across azimuth, elevation, and camera-to-object distance (Shen et al., 2023).

Consequently, it remains difficult to determine whether viewpoint-related performance differences arise from broader geometric exposure during training, from the quality of a particular observation viewpoint, or from access to complementary observations at inference time. This thesis addresses the gap by treating viewpoint as an explicit experimental variable within a controlled, object-centric synthetic benchmark. Three roles of viewpoint are disentangled: (i) its contribution to geometric diversity during training, (ii) its effect on the sensitivity of a fixed detector at inference, and (iii) the value of combining multiple viewpoint observations. Real-image transfer and adaptation experiments then establish

the boundary of external validity for these controlled synthetic findings.

Synthetic data is used here as an experimental instrument rather than a replacement for real UAV imagery. Viewpoint is parameterized by azimuth, elevation, and camera-to-object distance, with the dataset generated in Unreal Engine (Buck et al., 2022) and AirSim (Shah et al., 2018) to provide repeatable camera control and automated annotations. The synthetic train, validation, and test partitions are image-disjoint but not instance-disjoint: exact rendered images are not reused across partitions, while the same synthetic object instances and viewpoint conditions may occur in more than one split. This allows viewpoint to be studied under controlled conditions but limits how broadly the resulting patterns generalize. Simulated imagery also cannot fully reproduce the texture, sensor characteristics, background variability, illumination, and intra-class diversity of natural UAV footage (Eker et al., 2023). The thesis additionally evaluates the strongest synthetic detector, both zero-shot and after limited real-image fine-tuning, on public UAV imagery, same-taxonomy real imagery, and self-collected UAV data. These real-image experiments are used to test transfer and adaptability of the controlled synthetic findings rather than to validate the exact factor-level viewpoint effects observed under simulation.

1.1 Research Questions

The overarching research question is:

How does controlled viewpoint diversity shape detector learning, fixed-detector viewpoint sensitivity, and inference-time multiview performance in synthetic UAV object detection, and what boundary does real-world transfer reveal?

This question is divided into four experimental subquestions:

- 1: How does training viewpoint diversity affect held-out-viewpoint detector performance for known synthetic object instances on a fixed synthetic UAV benchmark?
- 2: Which viewpoint factors, namely azimuth, elevation, and camera-to-object distance, most strongly influence detection quality when the detector is fixed?
- 3: How does inference-time multiview availability affect target recovery, and are the resulting gains mainly explained by best-view availability or by explicit late fusion?
- 4: How well does the strongest synthetic detector transfer zero-shot to public UAV, same-taxonomy real-image, and self-collected real-drone datasets, and to what extent can limited real-image fine-tuning recover detection performance on held-out real UAV imagery?

Contributions. The contributions of this thesis are fourfold. (i) It constructs a controlled synthetic UAV benchmark in which object-centric viewpoint is parameterized by azimuth, elevation, and camera-to-object distance. (ii) It evaluates how restricted and broad viewpoint-exposure regimes affect detector learning, including equal-budget controls that separate geometric coverage from training-set size, and demonstrates that viewpoint utility for training-data selection is not equivalent to viewpoint quality during inference. (iii) It analyzes fixed-detector viewpoint sensitivity and the informational value of additional

inference-time observations under a target-centric matched-evidence protocol. This analysis separates cases where a single well-chosen viewpoint is sufficient to recover a correct detection from cases where accuracy improves gradually as more observations accumulate, and quantifies how marginal value changes as additional observations become available. (iv) The thesis evaluates the external boundary of the synthetic benchmark through zero-shot transfer and real-image fine-tuning.

Chapter 2

Background and Related Work

2.1 Object Detection Foundations

Object detection provides the technical foundation for the UAV based visual recognition task studied in this thesis. The task combines object localization in an image with assigning each localized instance to a semantic category. In UAV imagery, this computer vision task is strongly shaped by detector design. Proposal quality, feature resolution, and inference speed all influence whether small or weakly textured aerial targets survive the detection pipeline (Hua and Chen, 2025). Recent UAV surveys therefore discuss aerial detection not only in terms of overall accuracy, but also in terms of how two-stage, one-stage, and transformer-based detectors trade localization quality against efficiency under aerial constraints (Tang et al., 2024). These detector families build on a broader history of object detection methods, ranging from handcrafted-feature systems such as boosted cascades (Viola and Jones, 2001), histograms of oriented gradients (Dalal and Triggs, 2005), and deformable part models (Felzenszwalb et al., 2010). The limitations of earlier methods under scale variation, clutter, occlusion, and pose change are especially relevant in UAV imagery, where target appearance can vary substantially with camera height, distance, and bearing (Wu et al., 2019).

Two-stage detectors first generate candidate-region proposals and then classify and refine those proposals. Region-based convolutional networks first demonstrated that convolutional features could be combined with region proposals to achieve strong detection performance (Girshick et al., 2014). Fast R-CNN improved this pipeline by sharing convolutional computation across proposals (Girshick, 2015), while Faster R-CNN integrated proposal generation into the network itself through the region proposal network (Ren et al., 2017). These methods represent a proposal-based detection paradigm in which localization quality can be strong, but computational cost may be higher than before. This trade-off matters for UAV applications, where detectors may need to operate under onboard computation limits, communication constraints, or near-real-time requirements.

One-stage detectors provide a contrasting detection model by predicting bounding boxes and class probabilities in a single forward pass (Liu et al., 2016), while YOLO further popularized the idea of treating detection as direct regression from image regions to bounding boxes and class probabilities (Redmon et al., 2016). In UAV object detection, YOLO-based models are widely used for their practical balance between accuracy and inference speed, especially in settings involving small objects, scale variation, and resource-constrained deployment (Zhu et al., 2020). Many UAV-specific detectors retain a YOLO-style one-stage design while adapting it to aerial imagery through modified detection heads (Zhang et al., 2025), attention mechanisms (Zhao and Chen, 2024), refinement modules

(Chen et al., 2019), or multi-scale feature fusion (Hu et al., 2025).

Transformer-based detectors form a more recent detection family, but they are not the empirical focus of this thesis. DETR reframed object detection as a set-prediction problem and used attention mechanisms to model global image context (Carion et al., 2020). Such context modeling may be useful in UAV imagery, where small objects can depend on surrounding scene information. However, UAV surveys continue to emphasize that small-object handling, feature resolution, and computational efficiency remain central practical constraints in aerial detection (Hua and Chen, 2025). For this reason, transformer-based detectors are treated here as part of the broader detector landscape, while the experiments focus on established one-stage and two-stage detector families.

2.2 UAV Object Detection and Datasets

2.2.1 Unique Challenges in UAV Object Detection

UAV imagery differs from conventional ground-level object-detection imagery in scale, geometry, and context. Compared with benchmarks such as COCO (Lin et al., 2014), aerial targets often occupy fewer pixels, apparent object scale changes with altitude and standoff distance, and backgrounds contain structured clutter such as roads, rooftops, vegetation, and terrain (Zhu et al., 2021a). Scene density, background complexity, and occlusion are therefore treated as central difficulty factors in aerial detection rather than exceptional cases (Hua and Chen, 2025).

Camera-object geometry introduces an additional source of appearance variation. The same physical object can expose different surfaces, contours, and proportions when observed from another angle. This relation is commonly formalized through object pose. PASCAL3D+ parameterizes pose relative to the camera using quantities including azimuth, elevation, and distance (Xiang et al., 2014). The present thesis adopts this object-centric interpretation of viewpoint: viewpoint is treated as a structured relation between camera and target rather than as incidental image variation.

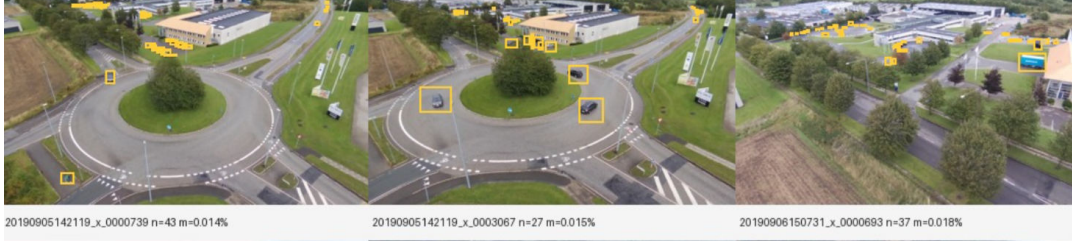
Rendered data provide one way to sample this relation systematically. Su et al. (2015) show that synthetic imagery with accurate viewpoint labels can be generated at scale for fine-grained viewpoint learning. The same principle is particularly relevant to UAV imagery because camera position can be actively changed through flight path, altitude, standoff distance, and orientation. Viewpoint is therefore not only a property of the recorded image; it can also be considered an experimental or operational variable.

2.2.2 UAV Datasets and Viewpoint Representation

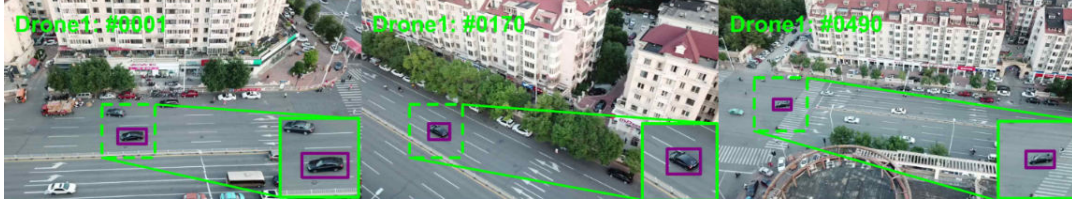
Public UAV benchmarks differ substantially in how camera viewpoint is represented or controlled. General real-world benchmarks such as UAVDT (Du et al., 2018), VisDrone (Zhu et al., 2021a), and AU-AIR (Bozcan and Kayacan, 2020) provide realistic variation in scale, altitude, motion, illumination, and occlusion, but camera angle typically varies together with object identity, background, and lighting (Tang et al., 2024). Consequently, viewpoint cannot easily be isolated.

Multiview datasets provide another form of viewpoint diversity. MAVREC (Dutta et al., 2023) and MDOT (Zhu et al., 2021b), for example, provide observations from multiple aerial platforms. This makes complementary observations of a target available, but the resulting camera-object geometries follow from the recording configuration rather than from a predefined object-centric sampling scheme. Synthetic and viewpoint-aware datasets provide greater control. SynDrone (Rizzoli et al., 2023) and UEMM-Air (Yao

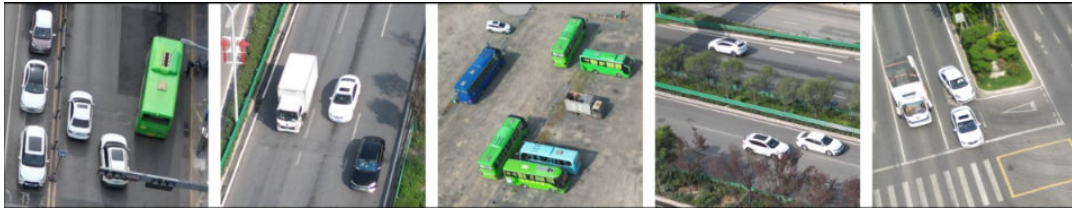
et al., 2025) increase the range of controllable aerial observations through simulation. Archangel adds UAV-position and pose metadata for human detection (Shen et al., 2023), while MAVSD controls discrete camera pitch angles (Li et al., 2025). MVPOD explicitly studies vertical perspective as a distribution shift (Jin et al., 2025). Similarly, Lee et al. (2022) use game-engine imagery to examine detector behaviour under changes in camera position, angle, illumination, and pose. Figure 2.1 shows representative examples of these different forms of viewpoint representation. Rather than serving as an exhaustive dataset comparison, the figure illustrates three progressively more structured settings: naturally occurring variation, multiview observation, and explicitly organized perspective variation.



(a) AU-AIR illustrates natural viewpoint variation in real UAV imagery. Camera geometry changes together with scene composition, target scale, background, illumination, and object identity, making the isolated effect of viewpoint difficult to determine.



(b) MDOT provides multiple UAV observations of the same target, enabling multiview analysis. The available views arise from the positions of multiple drones rather than from a predefined object-centric sampling of azimuth, elevation, and distance.

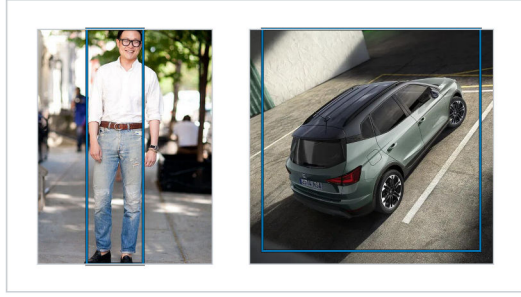


(c) MVPOD introduces a more explicit representation of viewpoint by distinguishing vertical perspectives, allowing detector performance to be studied across viewpoint-dependent distribution shifts. Its viewpoint organization remains different from dense object-centric sampling around a fixed target.

Figure 2.1: Examples of how existing UAV datasets represent viewpoint. **(a)** Real-world UAV benchmarks such as AU-AIR contain natural viewpoint variation that is entangled with other scene factors. **(b)** Multiview datasets such as MDOT provide several observations of the same target, but without a dense predefined camera–object pose lattice. **(c)** Viewpoint-aware datasets such as MVPOD make camera perspective more explicit, but do not construct the dense object-centric azimuth–elevation–distance sampling used in this thesis. Together, the examples illustrate the progression from naturally occurring viewpoint variation to increasingly explicit viewpoint representation.

2.2.3 From Natural Variation to Controlled Viewpoint Sampling

The datasets in 2.2.2 illustrate that viewpoint representation is not binary. Instead, existing work spans a continuum of geometric control. At one end, viewpoint occurs naturally as part of an uncontrolled scene. Multiview datasets make additional observations available but do not necessarily standardize the relation between camera and target. Viewpoint-aware datasets then impose greater structure on one or more camera parameters. For the present research question, however, observing different viewpoints is not sufficient by itself. To attribute changes in detector performance to viewpoint geometry, the same target must be observed repeatedly while the camera-object relation is varied in a predefined and reproducible way. Moreover, separating azimuth, elevation, and distance requires these variables to be represented independently rather than changing simultaneously with scene content or object identity. Figure 2.2 summarizes this progression conceptually. It first contrasts conventional object-detection imagery with the small-target and clutter conditions typical of UAV imagery, and then moves from naturally occurring viewpoint variation toward increasingly structured geometric sampling. The final panel represents the experimental design used in this thesis.



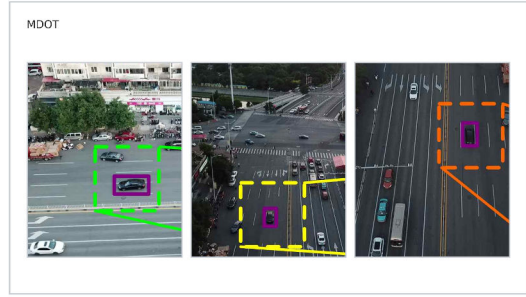
(a) **Conventional detection.** Target appearance, shape, and object boundaries are visually well resolved.



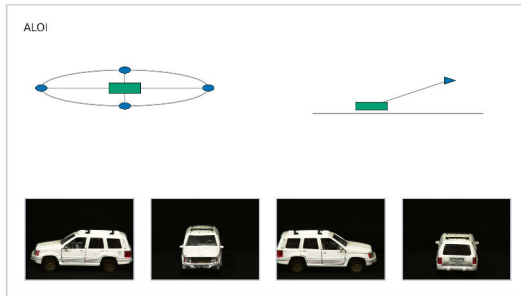
(b) **Real UAV detection.** Targets occupy only a small fraction of the AU-AIR image and must be detected within dense scene clutter. The inset illustrates the limited visual detail available for an individual target.



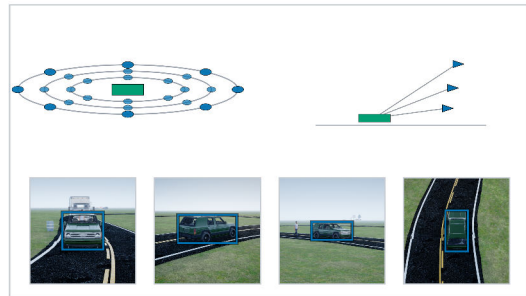
(c) **Natural viewpoint variation.** Camera geometry changes together with target scale, illumination, background, and scene composition, making the isolated effect of viewpoint difficult to determine.



(d) **Multiview availability.** Multiple observations of the same target are available, but their camera-object geometries arise from the recording configuration rather than from a pre-defined object-centric pose grid.



(e) **Partial viewpoint control.** Existing viewpoint-aware datasets introduce structured camera or perspective variation, but cover only part of the object-centric viewpoint space rather than densely sampling azimuth, elevation, and distance.



(f) **Dense object-centric control.** The dataset developed in this thesis systematically samples each target across eight azimuths, three elevations, and three camera-to-object distances, yielding 72 viewpoints per object.

Figure 2.2: Progression from conventional object-detection imagery to increasingly explicit representations of UAV viewpoint. Real UAV imagery introduces small targets and naturally entangled viewpoint variation; multiview and viewpoint-aware datasets provide progressively greater geometric structure; and the dataset developed in this thesis provides dense object-centric control over azimuth, elevation, and distance.

Table 2.1: Representative datasets.

Dataset	Viewpoint treatment	Viewpoint limitation
VisDrone & UAVDT (Zhu et al., 2021a; Du et al., 2018)	Realistic viewpoint variation through natural altitude, angle, and motion.	Viewpoint remains entangled with scene, identity, occlusion, and altitude; not controlled.
MVPOD (Jin et al., 2025)	Models vertical perspective explicitly as a distribution shift.	Focuses on domain shift rather than repeated object-centric sampling.
MAVSD (Li et al., 2025)	Controls discrete camera pitch angles.	Limited to pitch; no full azimuth–elevation–distance control.
Lee et al. (Lee et al., 2022)	Game-engine imagery with varied camera position, angle, illumination, and pose.	No dense object-centric viewpoint grid or separate training/sensitivity/inference analyses.
MDOT & MAVREC (Zhu et al., 2021b; Dutta et al., 2023)	Multiple platforms observe shared targets from different viewpoints.	Platform-centric; views are not systematically sampled around fixed targets.

2.3 Synthetic Data

Real-world aerial data collection is expensive and constrained by safety, privacy, logistics, weather, and annotation cost (Krump and Stütz, 2023). Synthetic imagery is often used simply as additional labeled training data. In this thesis, however, its value extends beyond data quantity: simulation allows viewpoint to be treated as an explicit and controlled experimental variable of unique quality.

Unreal Engine (Games, 2019) provides the virtual environment, while AirSim (Shah et al., 2018) enables programmable camera and UAV control together with pose information. This combination allows observations to be generated at predefined camera positions and compared under matched geometric conditions (Gui et al., 2024). Whereas viewpoint changes in real UAV data usually occur together with other environmental changes, simulation can render the same object instance from predefined azimuths, elevations, and distances while holding other scene properties constant (Dergachov and Aizatskyi, 2026).

Previous work demonstrates that rendered imagery can support detector training (Rajpura et al., 2017). Synthetic UAV datasets have similarly been used to increase aerial training data and scene diversity (Rizzoli et al., 2023). Controlled game-engine experiments can also be used to investigate detector behaviour under systematically modified imaging conditions (Lee et al., 2022). The present thesis primarily uses simulation in this latter role: as a controlled experimental instrument for measuring viewpoint effects.

This control makes it possible to distinguish mechanisms that are otherwise easily conflated. Broader viewpoint exposure during training may improve the range of appearances represented by a detector’s training distribution (Tobin et al., 2017). Separately, making additional views available during inference can increase the probability that at least one observation provides useful evidence. These mechanisms require different experimental comparisons and are therefore evaluated separately in this thesis.

2.3.1 Synthetic-to-Real Transfer

The experimental control provided by simulation also introduces an important limitation. Synthetic imagery can differ from real UAV imagery in texture, illumination, background statistics, sensor characteristics, and intra-class variation. Strong performance within a synthetic environment therefore does not by itself establish real-world detector competence.

Domain randomization addresses this problem by deliberately increasing variation in simulation so that real observations are more likely to fall within the visual range encountered during training (Tobin et al., 2017). Another strategy is to adapt a synthetic-trained detector using annotated real data. Prior work shows that synthetic imagery can be combined with real-data training or fine-tuning (Rajpura et al., 2017). Domain-randomized synthetic data have similarly been investigated as a means of reducing the amount of real annotation required (Tremblay et al., 2018). RarePlanes provides another example in which synthetic and real overhead imagery are evaluated jointly to investigate the contribution of simulation to aerial recognition (Shermeyer et al., 2021). Zero-shot transfer and real-image fine-tuning answer different questions.

Zero-shot evaluation measures direct generalization from simulation to the target domain without target-domain supervision. Fine-tuning instead measures whether the learned representation can be adapted once real labeled examples become available. This thesis uses zero-shot evaluation to establish the external-validity boundary of the controlled synthetic experiments and subsequently uses real-image fine-tuning to evaluate adaptability. Because the experimental design does not include a matched real-only training baseline, the fine-tuning experiment is interpreted as evidence that the synthetic-trained detector can be adapted to real UAV imagery. It is not interpreted as proof that synthetic pretraining itself improves final real-world performance relative to training on the same real data alone.

2.4 Multiview Observation and Fusion Strategies

An observation that is weak from one viewpoint may be more informative from another. This motivates multiview detection systems in which several cameras or platforms observe the same environment. Park et al. (2023) show that joint modeling of multiple views can improve detection by combining information across cameras. In an aerial context, Nathan et al. (2023) show in simulation that coordinated drone sampling can recover target visibility that is missed under less informed sampling strategies. These findings motivate an important distinction for the present thesis: additional viewpoints can provide value even when the underlying detector is kept fixed. A second observation may simply provide a more favorable camera-object geometry than the first. Alternatively, several observations may contain complementary evidence that becomes stronger when combined. Separating these mechanisms is central to the inference-time analysis.

2.4.1 Fusion Strategies

Multiview information can be combined at different stages of a recognition pipeline. Early fusion combines raw or minimally processed inputs, intermediate fusion combines learned feature representations, and late fusion combines model outputs or decisions (Baltrušaitis et al., 2019). Intermediate representations can also be combined through dedicated multiview architectures (Yazgan et al., 2024). The present thesis uses fixed-detector late fusion. Single-view detector predictions are generated first, after which predictions from different observations are compared or combined. This allows inference-time viewpoint

effects to be studied without retraining a dedicated multiview network and without assuming that every camera shares a calibrated feature representation.

Within late fusion, two conceptually different strategies are relevant. Selection-based methods retain the strongest available observation. A best-view or maximum rule therefore measures whether an additional viewpoint can *rescue* a target that was poorly observed in another view. This is related to max-style aggregation used in multiple-instance learning (Ilse et al., 2018).

Evidence-accumulation methods instead combine information from several observations. At the score level, several weak-to-moderate observations can jointly increase target support without requiring all boxes to occupy a common image plane (Daryani et al., 2025). At the box level, detections can be merged or reweighted using approaches such as Soft-NMS (Bodla et al., 2017) or Weighted Boxes Fusion (Solovyev et al., 2021). Such box-level methods require detections to be meaningfully comparable geometrically, which may require valid reprojection in a multiview UAV setting.

2.4.2 Coalition-Based Viewpoint Analysis

A multiview observation set can also be interpreted as a coalition in which each viewpoint contributes to the quality of the complete set. Coalition analysis makes it possible to ask not only whether additional views improve performance, but also which views contribute most to that improvement.

Shapley values provide a principled method for distributing a coalition value among individual contributors (Shapley, 1953). Each viewpoint is treated as a player, and its contribution is determined from its marginal effect when added to different predecessor coalitions. In the present context, the coalition value can represent the strongest available target evidence or the quality obtained after combining evidence across observations. This formulation is useful because the contribution of a viewpoint need not be equivalent to its standalone quality. A view that is individually strong may add little when similar evidence is already present, whereas a weaker but complementary view may contribute substantially when it rescues cases missed by the existing coalition.

2.4.3 Separating Mechanisms of Multiview Gain

The term viewpoint diversity can refer to several distinct mechanisms. First, a detector may encounter a broader range of camera-object geometries during training. MVPOD shows that mismatches between training and test perspectives can substantially affect detector performance (Jin et al., 2025), motivating the training-side viewpoint-diversity analysis in this thesis. Second, additional viewpoints may become available only at inference time, increasing the chance that at least one observation provides a favorable view of the target. This selection or rescue effect does not require evidence to be fused. Third, multiple observations may provide complementary evidence whose combined value exceeds that of the strongest individual view (Hou et al., 2020).

These mechanisms are not interchangeable. An inference-time gain, for example, does not necessarily imply successful evidence accumulation if the same improvement can be obtained through best-view selection. Moreover, fixed-camera multiview systems may not generalize directly to changing camera counts or geometries (Vora et al., 2023).

2.5 Research Gap

Existing UAV datasets provide realism, multiview coverage, or partial viewpoint control, but these properties are rarely combined within a controlled object-centric framework. Real-world datasets do not generally isolate camera-object geometry, while multiview datasets obtain viewpoints from their recording configuration. Viewpoint-aware and synthetic datasets provide greater geometric structure, but typically control only part of the viewpoint space or study viewpoint as one factor within a broader task. Consequently, the literature does not provide dense repeated sampling of the same targets across azimuth, elevation, and distance while separately examining the mechanisms through which viewpoint affects detection.

This thesis addresses this gap through four linked experiments: (i) *training-side viewpoint diversity*, examining how training coverage relates to held-out-viewpoint performance; (ii) *fixed-detector viewpoint sensitivity*, measuring how performance varies with azimuth, elevation, and distance; (iii) *inference-time multiview analysis*, separating best-view rescue from evidence accumulation; and (iv) *real-world transfer and adaptation*, testing the external-validity boundary of the synthetic findings through zero-shot transfer and real-image fine-tuning.

The first three experiments use synthetic data primarily for geometric control, while the fourth examines how these findings transfer toward real UAV imagery.

Chapter 3

Methodology

The experimental pipeline (Figure 3.1) was designed around a controlled synthetic capture space in which viewpoint could be varied systematically while object identity, scene structure, and evaluation protocol remained fixed. Four linked experiments use this shared azimuth-elevation-radius grid and standardized evaluation outputs to examine, in turn, training-side viewpoint diversity, fixed-detector viewpoint sensitivity, inference-time multiview availability, and real-world transfer and adaptation.

3.1 Overview of the Experimental Design

The methodology proceeds in four stages, each answering one research question.

1. **Training-side viewpoint diversity** (Experiment 1, RQ1): three detector families are trained under five viewpoint-exposure regimes (M1–M4, Table 3.3) and evaluated on a common held-out test set, isolating the effect of broader training-viewpoint coverage from training-set size via equal-budget controls.
2. **Fixed-detector viewpoint sensitivity** (Experiment 2, RQ2): the strongest detector from Experiment 1 is analyzed across the full viewpoint grid to determine which of azimuth, elevation, and radius drives detection quality when training exposure is held constant.
3. **Inference-time multiview analysis** (Experiment 3, RQ3): cached predictions from the fixed, strongest detector are used to simulate access to one, two, or three viewpoints per target, separating rescue availability from evidence accumulation without any further training.
4. **Real-world transfer, adaptation, and exploratory same-object analysis** (Experiment 4, RQ4): the strongest detector is evaluated zero-shot on public and self-collected real UAV imagery, then fine-tuned on a small labelled real subset, with a final exploratory multiview analysis on five physical targets photographed from multiple real viewpoints.

Because the detector is kept fixed within Experiments 2 and 3, and training exposure is the only manipulated variable in Experiment 1, effects attributable to training exposure, observation geometry, and inference-time view availability can be examined separately rather than conflated.

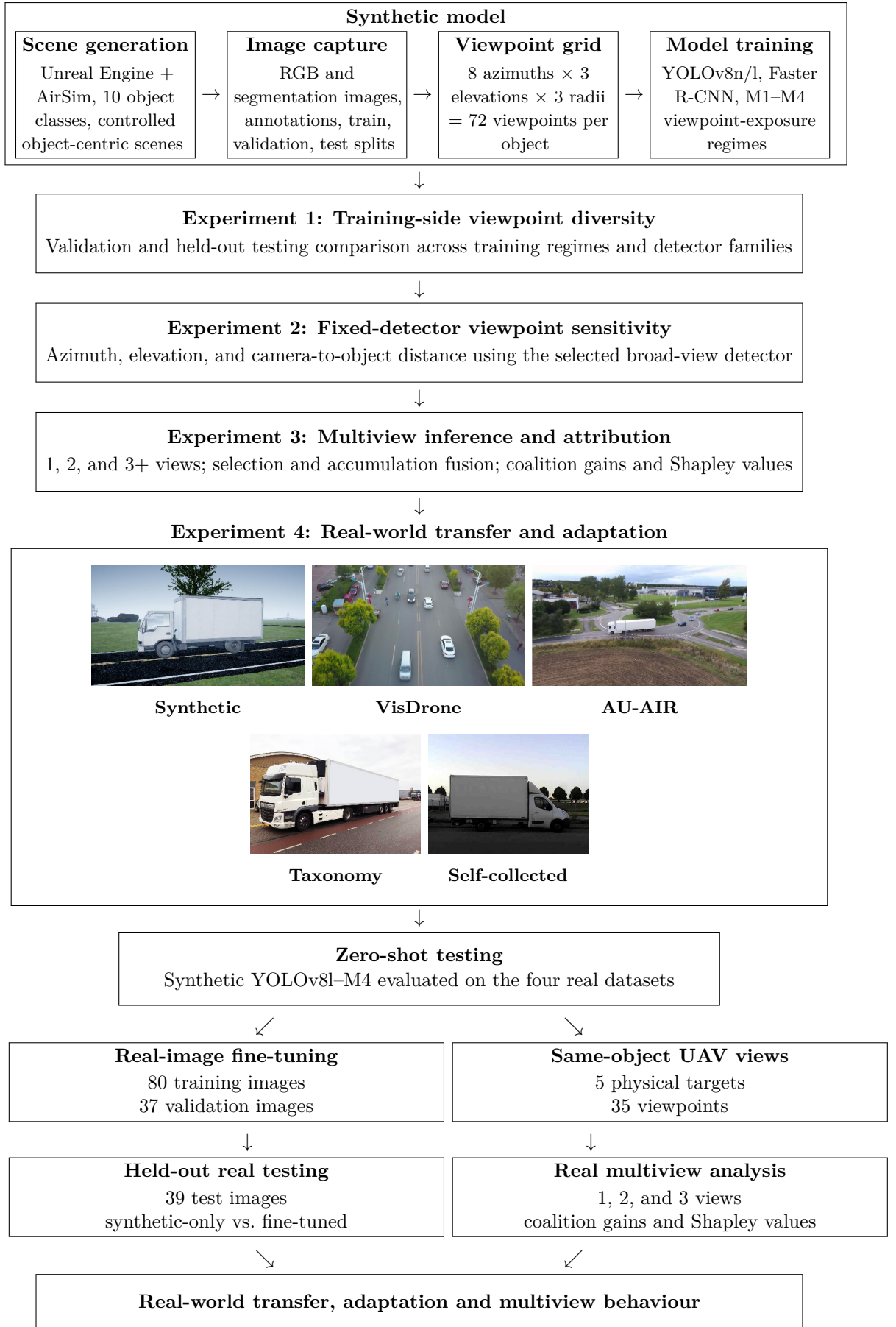


Figure 3.1: Overview of the experimental pipeline, from synthetic scene generation and viewpoint sampling to detector training and controlled viewpoint analyses, followed by zero-shot real-world transfer, real-image fine-tuning, held-out testing, and multiview evaluation.

3.2 Synthetic Environment and Dataset Generation

The synthetic dataset is used as a controlled experimental instrument rather than as a direct substitute for real UAV imagery. It serves three complementary purposes. (i) It allows camera–object geometry to be varied systematically by repeating the same object-centric azimuth–elevation–radius capture pattern under matched conditions. (ii) The rendered object assets provide repeated examples of class-specific morphology, including silhouette, proportions, edges, and visible surface structure across viewpoints, while omitting part of the uncontrolled appearance variation present in real UAV imagery, such as sensor noise, natural background diversity, weather, and scene-specific texture. (iii) The synthetic pipeline provides automatically annotated training data at substantially greater scale than would be practical to collect manually. Because broader viewpoint coverage can therefore coincide with a larger number of training images, later equal-budget experiments explicitly separate the effect of viewpoint diversity from the effect of training-set size. The benchmark is implemented in Unreal Engine with automated annotations derived from rendered outputs, prioritizing experimental control and internal validity over direct photorealistic equivalence to real UAV imagery; scene appearance is therefore deliberately constrained during the controlled viewpoint experiments.

3.2.1 Object Classes

The dataset contains ten object classes (*tent*, *tank*, *tower*, *container*, *whitevan*, *suv*, *male*, *rock*, *barrel*, *tree*), selected to span differences in geometry, size, texture, silhouette stability, and expected viewpoint sensitivity rather than to reproduce any single UAV benchmark taxonomy. The synthetic assets expose the detector to class-specific morphological cues such as elongated vehicle bodies, box-like containers, narrow vertical structures, rounded or irregular objects, and articulated human shape. These cues remain informative across viewpoints even though the rendered imagery does not reproduce the full texture, noise, illumination variability, and environmental complexity of real UAV data.

Person- and vehicle-related categories preserve comparability with Table A.1, while larger rigid structures (*tank*, *container*, *tower*), a larger low-texture structure (*tent*), and smaller or clutter-like objects (*barrel*, *rock*, *tree*) follow prior guidance that aerial datasets should vary object size and scene structure rather than only image count (Hua and Chen, 2025; Krump and Stütz, 2023). Each target object is placed in a controlled simulated environment, with the camera moving around it according to the viewpoint grid in Figure 3.2. The 3D assets are obtained from Fab. Terrain, sky, and lighting are intentionally kept simple and stable across the controlled viewpoint captures. This is a deliberate internal-validity choice: keeping scene appearance fixed prevents background or illumination changes from becoming systematically associated with particular viewpoints, allowing differences between matched observations to be related more directly to camera–object geometry. The resulting reduction in environmental diversity is treated as a limitation of the synthetic benchmark and is examined further through the real-world transfer experiments.

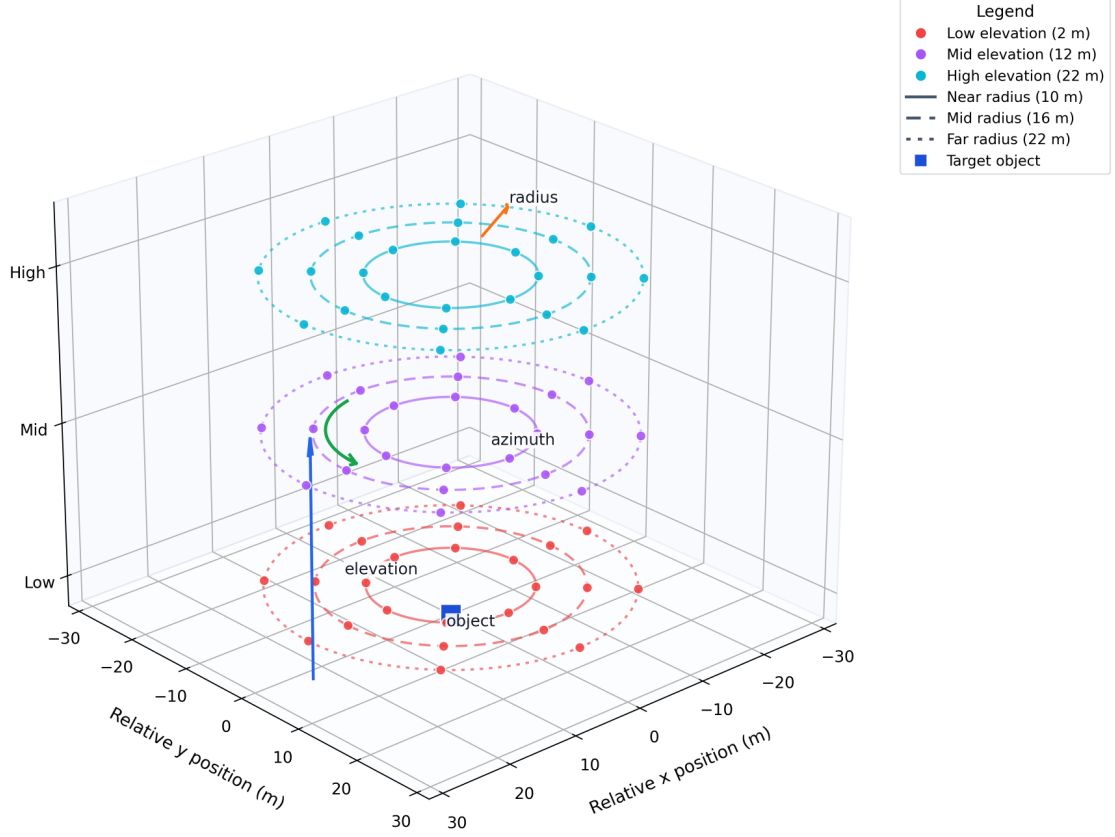


Figure 3.2: The deterministic 72-view capture grid: eight azimuths, three radius shells, and three vertical-offset elevation bands per object instance.

3.2.2 Camera-object geometry

Each object instance is captured on a deterministic $8 \times 3 \times 3 = 72$ viewpoint grid: eight azimuths, three vertical-offset elevation bands, and three radii (Table 3.1). This decouples three viewpoint components often entangled in real UAV imagery, horizontal bearing, vertical viewing geometry, and camera-to-object distance (Wu et al., 2019), consistent with the standard azimuth-elevation-distance pose descriptor (Xiang et al., 2014). Azimuth is the horizontal bearing around the target in equal 45° steps, giving full coverage while remaining dense enough to distinguish frontal, side, rear, and diagonal views without attempting to estimate continuous pose (Su et al., 2015). The elevation band is the camera’s vertical offset above the target (2 m / 12 m / 22 m), described throughout as *low*, *mid*, and *high* camera height above target. This range is deliberately bounded, wide enough to vary visible surface structure and self-occlusion, narrow enough to avoid degenerating into either a no-variation or a small-object study. Radius is the horizontal camera-to-object distance (10 m / 16 m / 22 m), controlling apparent scale (Hua and Chen, 2025) while keeping the target visible across all conditions, so that quality differences reflect viewpoint geometry rather than target disappearance. The exact coordinate parameterization is given in Appendix A.2.1.

Table 3.1: Deterministic viewpoint grid used for each object instance.

Factor	Levels
Azimuth	$0^\circ, 45^\circ, 90^\circ, 135^\circ, 180^\circ, 225^\circ, 270^\circ, 315^\circ$ (full 360° coverage).
Elevation band	Low (2 m), Mid (12 m), High (22 m) camera height above target.
Radius	Near (10 m), Mid (16 m), Far (22 m) camera-to-object distance.

3.3 Image Capture, Annotation and Dataset Audit

Aligned RGB and segmentation outputs are rendered at each predefined camera pose via AirSim (Shah et al., 2018), from which bounding boxes are derived automatically, avoiding manual annotation noise and enabling exact repetition of viewpoint configurations across objects. Figure 3.3 shows representative examples. Figure 3.4 summarizes the pipeline used in this thesis.

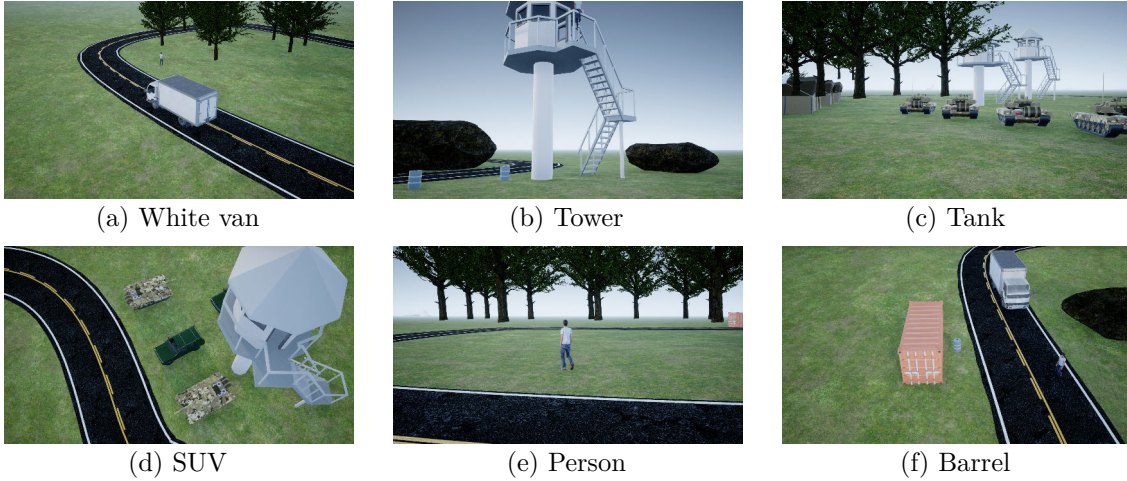


Figure 3.3: Representative author-generated images from the synthetic UAV dataset, rendered in Unreal Engine and captured through the AirSim pipeline. The examples illustrate variation in object class, apparent scale, scene context, and camera-object geometry across azimuth, elevation, and distance.

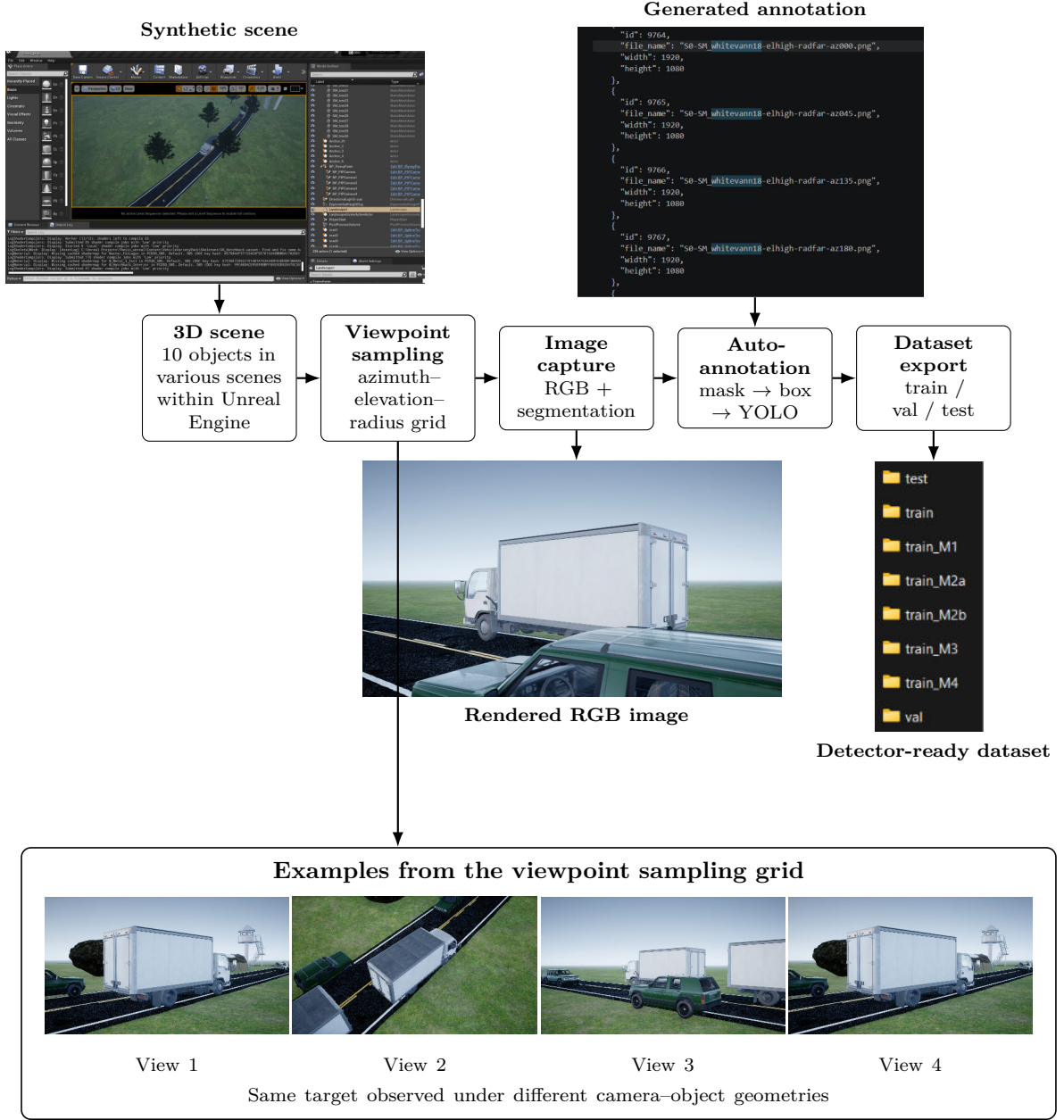


Figure 3.4: Synthetic dataset-generation pipeline. A target scene is instantiated in Unreal Engine and sampled using a structured azimuth-elevation-radius viewpoint grid. AirSim produces RGB and segmentation outputs, from which object-detection annotations are generated automatically before the images and labels are exported into train, validation, and test splits. The lower panel illustrates the appearance of the same target under different sampled viewpoints.

3.3.1 Dataset Audit and Split Structure

A dataset audit verified the generated splits: 14,760 total images (10,332 train / 2,214 validation / 2,214 test) across 205 inferred unique object instances, with all ten classes represented in every split. The dataset is not perfectly balanced (*tree* provides the most images and instances, *tower* the fewest), so class-level results should be interpreted together with their available support. The full per-class breakdown is given in Appendix A.2.1.

3.3.2 Annotation Format and Target-Box Selection

Each rendered image has a YOLO label file giving, per visible object, a class identifier and a normalized bounding box; when multiple boxes of the same class appear, the intended focus object is recovered from a per-capture manifest by selecting the box of the focus class closest to the image center. The exact pixel-coordinate conversion and box-selection criterion are given in Appendix A.2.1.

3.4 Detection Models and Training Protocol

The comparison spans two one-stage detectors and one two-stage detector, testing whether viewpoint-diversity effects are specific to one detection paradigm. YOLOv8n (Ultralytics, 2023; Jocher et al., 2023) provides a lightweight, efficiency-oriented one-stage baseline; YOLOv8l shares this paradigm with substantially greater capacity, testing whether capacity improves generalization under viewpoint variation. Faster R-CNN (Ren et al., 2017) provides a two-stage comparison point via its region-proposal-network design, allowing viewpoint-related patterns to be checked across fundamentally different detection paradigms. The training protocol is held fixed within each detector family across all five regimes, so the intended experimental variable is the training-viewpoint subset. Some implementation differences remain between families due to differing frameworks, preprocessing, and schedules. Full hyperparameters are archived in Appendix A.2.1. Figure 3.5 compares YOLOv8l and Faster R-CNN predictions on the same representative M4 test frame, using cached repository predictions to illustrate the qualitative detection behaviour that recurs in the benchmark results.

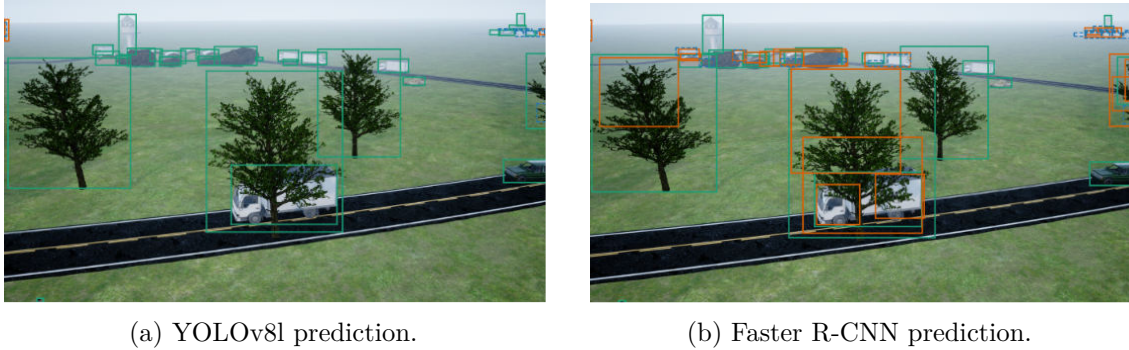


Figure 3.5: Repository-native comparison of YOLOv8l and Faster R-CNN predictions on the same representative M4 test frame. Panel (a) shows YOLOv8l and panel (b) shows Faster R-CNN.

3.5 Evaluation Metrics

All trained detectors are evaluated on the same test split using one standardized pipeline. Precision, recall, and F1 are computed from one-to-one class-wise box matching (equations in Appendix A.2.1). Average precision (AP) summarizes the precision–recall curve obtained by varying the confidence threshold, following the COCO convention (Lin et al., 2014). At a given intersection-over-union (IoU) threshold t , AP_t measures detection performance when a prediction is considered correctly localized only if its IoU with the matched ground-truth box is at least t . The primary model-ranking metric in this thesis is $mAP_{50 : 95}$,

which averages AP over the ten IoU thresholds from 0.50 to 0.95 in steps of 0.05:

$$\text{mAP}_{50 : 95} = \frac{1}{10} \sum_{t \in \{0.50, 0.55, \dots, 0.95\}} \text{AP}_t. \quad (3.1)$$

By contrast, AP50 evaluates performance only at $t = 0.50$. The stricter $\text{mAP}_{50 : 95}$ metric therefore penalizes poorly localized detections more strongly and is used as the primary model-ranking metric throughout the synthetic detector experiments.

Three related, but distinct, scores recur throughout the thesis and are summarized in Table 3.2.

Table 3.2: Scores used throughout the thesis and their role.

Term	Used for
Dataset-level $\text{mAP}_{50 : 95}$	Ranking complete trained detectors (Experiment 1).
Per-image $\text{AP}_{50 : 95}$ proxy	Comparing detection quality across viewpoint conditions (Experiment 2).
Strict target quality ($q = c \times \text{IoU}$)	Target-centric multiview comparisons (Experiment 3, Experiment 4).

The strict target-quality score $q(u) = c(u) i(u)$, the product of matched confidence and matched IoU for a single view u , is defined here because it is specific to this thesis’s multiview analyses (§3.8) and is used as the headline metric wherever fusion policies are compared, since it is the only score well-defined for every coalition family.

Where results are aggregated across repeated experimental units, the mean is reported together with its standard error (SE),

$$\text{SE} = \frac{s}{\sqrt{n}}, \quad (3.2)$$

where s is the sample standard deviation and n is the number of independent units contributing to the reported mean. The aggregation unit depends on the experiment. For the single- and pair-view training sweeps, uncertainty describes variation across the evaluated viewpoint configurations; for the synthetic multiview analysis, SE is computed across scene-level means; and for the real same-object case study, SE is computed across physical-object-level means. For the training-viewpoint sweeps, these standard errors characterize variability across viewpoint configurations rather than run-to-run optimization uncertainty, since each configuration was trained once. Similarly, metrics reported for a single trained checkpoint on a fixed test set are treated as dataset-level evaluation statistics and are not assigned an SE unless an explicit aggregation or resampling procedure is used.

3.6 Experiment 1: Training-Side Viewpoint Diversity

Experiment 1 tests whether broader geometric coverage during training improves generalization to held-out viewpoints. Five training regimes are compared, ranging from a single canonical viewpoint (M1) to full joint azimuth–elevation–radius coverage (M4), as summarized in Table 3.3. All regimes are evaluated on the same held-out test split. Because broader regimes also contain more images, the main regime comparison alone

is confounded by training-set size, a limitation addressed directly by the equal-budget controls below. The partitions are image-disjoint but not instance-disjoint. Exact rendered images are not reused across partitions, while the same inferred synthetic object instances may occur in more than one partition. The benchmark therefore evaluates performance on held-out rendered observations of recurring synthetic instances rather than generalization to unseen object identities.

Two supporting sweeps, training one YOLOv8l detector per individual viewpoint cell, and per unordered pair of cells, characterize how training utility varies across the viewpoint lattice; their full procedures are given in Appendix A.2.2.

Table 3.3: Notation for the five viewpoint-exposure training regimes. Exact viewpoint cells and image counts are given in Appendix A.2.1.

Regime	Training exposure	Purpose
M1	Canonical single-view training.	Tests generalization from one standard viewpoint to the full viewpoint distribution.
M2a	Elevation variation at fixed azimuth and radius.	Tests whether vertical viewpoint diversity alone improves generalization.
M2b	Limited azimuth variation across elevation bands.	Tests partial horizontal and vertical viewpoint coverage.
M3	Full azimuth variation at fixed elevation and radius.	Tests whether horizontal viewpoint diversity alone is sufficient.
M4	Full azimuth, elevation, and radius variation.	Complete multiview training regime; main baseline for later viewpoint analyses.

3.6.1 Equal-Budget Controls

The regime comparison and the single-/pair-view sweeps are all confounded by training-set size, a broader-coverage condition simply uses more images. The equal-budget controls resolves this by matching the training-image count of a restricted single- or pair-view condition, while drawing those images from across the full M4 viewpoint space rather than from one or two fixed cells. If the restricted condition still outperforms its equal-budget control, the specific viewpoints are useful beyond image count alone; if the equal-budget control performs better, broad viewpoint diversity is more valuable than concentrating the same image budget in few viewpoints. This comparison directly supports the thesis’s main training-side claim by isolating viewpoint coverage from image count.

3.7 Experiment 2: Fixed-Detector Viewpoint Sensitivity

Experiment 2 asks which components of camera-object geometry remain influential once the detector itself is fixed. With the strongest detector from Experiment 1 (YOLOv8l-M4) held constant, an exact-cell analysis evaluates each azimuth-elevation-radius combination separately. A complementary factor-level analysis aggregates across one factor at a time to determine whether systematic variation is strongly associated with azimuth, elevation, or radius. For each class, the dominant viewpoint factor is the one with the largest best-minus-worst range across its levels. This separates observation-dependent detector sensitivity from the training-side effects examined in Experiment 1.

A supporting clutter-stratification check (Appendix A.2.5) confirmed that the target-centric benchmark retains meaningful variation in detection difficulty. Since clutter is not itself a component of camera-object viewpoint, this analysis does not directly answer RQ2 and serves only to characterize benchmark difficulty.

3.7.1 Viewpoint-Dependence Analysis

With the detector fixed as YOLOv8l-M4, variation across the viewpoint grid reflects observation-dependent detector behaviour rather than training-regime differences. An exact-cell analysis evaluates each azimuth-elevation-radius combination separately. A factor-level analysis aggregates across one factor at a time to determine whether systematic variation is most strongly associated with azimuth, elevation, or radius. The factor-level score for class c , factor f , and level ℓ is

$$\bar{s}_{c,f,\ell} = \frac{1}{N_{c,f,\ell}} \sum_{i \in \mathcal{I}_{c,f,\ell}} s_i,$$

where s_i is the cached per-image AP50 : 95 proxy and $\mathcal{I}_{c,f,\ell}$ the relevant image set. For each class, the dominant viewpoint factor is the one with the largest best-minus-worst range across its levels.

3.8 Experiment 3: Inference-Time Multiview Analysis

Experiment 3 asks what additional observations contribute when the detector is kept fixed and no further training occurs. This isolates the effect of additional observations at inference time from the effect of additional training data, and separates three sources of value: *rescue availability* (an additional observation succeeds when another is weak), *evidence accumulation* (multiple matched observations provide value beyond selecting the strongest single detection), and *diminishing marginal value* (how quickly the benefit of an additional view decreases as more observations become available). We refer to this analysis throughout as a target-centric multiview diagnostic, since it uses ground-truth IoU for matching and assumes target correspondence across views.

3.8.1 Fixed-Detector Operational Multiview Protocols

For each view u , target quality is defined as

$$q(u) = c(u)i(u),$$

where $c(u)$ is the confidence of the detection matched to the target at IoU50, and $i(u)$ is its IoU with the ground-truth box. Since IoU requires ground truth, $q(u)$ is an evaluation-time diagnostic rather than a score available during deployment.

For a set of available views S , the scores are ordered as $q_{(1)}(S) \geq \dots \geq q_{(|S|)}(S)$, and the coalition value is

$$v_r(S) = q_{(r)}(S),$$

where r specifies the required level of multiview support. The $r = 1$ rule selects the strongest available view, $v_1(S) = \max_{u \in S} q(u)$, while for three-view coalitions $r = 2$ uses the second-best score and $r = 3$ the weakest score. These correspond to requiring strong evidence from at least one, at least two, or all three views, respectively. The same ranking rules are applied to the per-image quality proxy and to binary target-detected indicators.

3.8.2 Matched-Box Fusion Policies

This analysis remains target-centric and keeps the detector fixed, operating on matched target boxes. Three policies anchor the comparison: *best box* (selects the single strongest matched observation, a pure selection baseline testing rescue availability), *support-weighted Noisy-OR* (combines confidence with cross-view support, rewarding corroboration and penalizing incomplete support), and *Noisy-OR + best IoU* (tests whether accumulating evidence across views improves target quality beyond selecting the best individual observation). Conceptually, best box asks only "was the target seen well from at least one view?", while the Noisy-OR variants ask "does agreement across views raise our confidence beyond the best single view?" Additional reference variants (mean quality, unanimous best box, Noisy-OR with mean IoU, odds-product corroboration), the full threshold-family definitions, the detailed challenge-slice construction, and the complete Shapley formulation with its numerical verification procedure are given in Appendix A.2.5 and Appendix A.2.6. In brief, coalition analysis (Shapley values, (Shapley, 1953)) is used to quantify how the marginal value of an additional observation changes as the coalition grows from one view to two, two to three, and beyond, without attributing value to specific angles or UAV agents.

3.9 Experiment 4: Real-World Transfer and Adaptation

Experiment 4 tests the external boundary of the controlled synthetic benchmark. The first stage evaluates whether the strongest synthetic detector (YOLOv8l-M4) transfers zero-shot to public UAV, same-taxonomy real, and self-collected real-UAV imagery, without target-domain supervision. The second stage tests whether the same checkpoint can be adapted once labelled real UAV images become available. These answer different questions: zero-shot transfer measures direct transfer, fine-tuning measures adaptability.

3.9.1 Zero-shot real-world Transfer

No real-image fine-tuning, domain adaptation, or class-specific recalibration is applied at this stage; the purpose is to measure how far the representation learned in the controlled Unreal Engine domain transfers directly to real or more naturalistic imagery. Four settings are evaluated: VisDrone and AU-AIR (using approximately overlapping class mappings), an internet-curated same-taxonomy dataset, and a self-collected UAV dataset built around the synthetic benchmark's ten classes. None of these datasets contains a complete repeated azimuth-elevation-radius grid around the same physical targets, so they test transfer rather than reproducing the controlled viewpoint analyses of Experiments 1–3. Figure 3.6 provides a qualitative comparison of the controlled synthetic benchmark and the real-image evaluation domains. Even when semantically similar objects are selected, substantial differences remain in texture, target scale, background complexity, illumination, image quality, and sensor characteristics.

3.9.2 Self-Collected Real-UAV Dataset Preparation

The self-collected dataset provides a real-domain counterpart to the ten-class synthetic taxonomy. Images were captured personally using a Loolinn U61 quadcopter drone equipped with a 90° adjustable onboard camera producing images at a resolution of 640×480 pixels. The drone connects to a smartphone over a dedicated Wi-Fi link, streaming live first-person-view footage to the companion app, through which the aircraft



(a) Synthetic benchmark (generated by the author using Unreal Engine and AirSim)



(b) AU-AIR (Bozcan and Kayacan, 2020)



(c) Internet same-taxonomy image (vevr-vrachtwagenreclame)



(d) Self-collected UAV data (author)

Figure 3.6: Representative appearance variation across the synthetic and real-image domains used in the transfer evaluation.

is piloted. Still images were captured rather than extracted from recorded video, as frame extraction was found to produce visibly lower image quality than photographs taken directly through the app. Photographs were taken at low altitude over outdoor scenes containing instances of the target classes, under naturally varying flight altitude, distance, and viewing angle rather than a controlled experimental protocol, to reflect realistic handheld UAV operation. This resulted in a final 156-image dataset representing the classes *tent*, *tank*, *tower*, *container*, *whitevan*, *suv*, *male*, *rock*, *barrel*, and *tree*. Because these images represent naturally varying real-world viewing conditions rather than the deterministic synthetic grid, they are used to evaluate domain transfer rather than to reproduce the controlled factor-level viewpoint analysis.



Figure 3.7: Real-UAV image acquisition setup. Still images for the self-collected dataset were captured using the pictured Loolinn U61 drone, piloted via a smartphone-connected FPV app.

3.9.3 Real-Image Fine-Tuning Protocol

The real-image adaptation experiment starts from the same YOLOv8l-M4 checkpoint identified as the strongest detector in the synthetic benchmark. Rather than initializing a

new detector, the weights learned under full synthetic viewpoint exposure are used as the starting point for fine-tuning on the 80-image real training split. This setup tests whether the synthetic-trained detector can be adapted toward the real UAV domain in a low-data real-image setting, rather than through large-scale real-domain retraining.

During fine-tuning, a separate 37-image validation split is used to monitor model performance and select the final checkpoint. The held-out 39-image test split is not used during optimization or checkpoint selection. After training, both the original synthetic-only YOLOv8l-M4 model and the fine-tuned model are evaluated independently on exactly the same 39 test images.

Fine-tuning was performed at an input resolution of 640×640 pixels for 60 epochs using a batch size of 8, the AdamW optimizer, and an initial learning rate of 0.0005. Figure 3.8 illustrates the within-object viewpoint variation used in the real multiview analysis. All three images depict the same physical truck, but its visible appearance changes substantially between the frontal, lateral, and elevated oblique viewpoints.

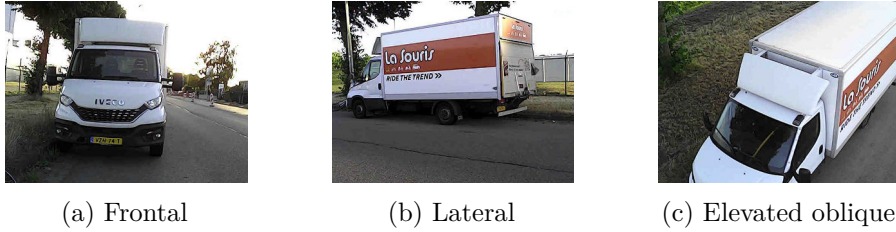


Figure 3.8: Author-acquired observations of the same La Souris truck using the self-collected UAV setup: (a) frontal, (b) lateral, and (c) elevated oblique views.

3.9.4 Fine-Tuning Evaluation

Both checkpoints are evaluated identically at confidence ≥ 0.25 and IoU ≥ 0.50 , from which precision, recall, F1, AP50, and mAP50 are computed. The 39-image test subset is strongly class-imbalanced (of 144 labelled objects, 84 are *suV*, 33 *whitevan*, 13 *male*, with the remaining seven classes contributing one to three instances each); results are therefore reported both as an all-ten-class summary and as a higher-support analysis restricted to classes with at least ten test instances (*whitevan*, *suV*, *male*), which provides the primary basis for interpreting adaptation performance, with sparsely represented classes treated as exploratory diagnostics only.

3.9.5 Real Same-Object Multiview and Exact Shapley Analysis

As an exploratory follow-up to the real-image fine-tuning check, five self-collected UAV sequences were used to examine whether repeated views of the same physical target provide additional detection value after domain adaptation. The sequences contain a parking-lot car (six views), the Ooij tower (seven views), the La Souris truck (ten views), the M truck (five views), and a white truck (seven views), giving 35 usable viewpoints across the *suV*, *tower*, and *whitevan* classes.

Coalitions were formed only within each physical object, so the analysis asks how much different views of the same target contribute at inference time rather than training a multiview detector. Each view was evaluated with both the synthetic YOLOv8l-M4 checkpoint and the real-image fine-tuned checkpoint. A target was counted as detected when a correct-class prediction matched the ground truth with IoU ≥ 0.50 and confidence

≥ 0.25 . For each view, strict target quality was defined as $q_i = c_i \cdot \text{IoU}_i$, with $q_i = 0$ when no valid match was found. Exact Shapley values were then computed separately for each object and model over all $2^{|N|}$ same-object view subsets. The primary coalition value was the maximum strict quality achieved by any view in the subset, while binary detection rescue was used as a secondary value function. Because the real sequences are small and not a full held-out multiview benchmark, the results are interpreted descriptively: they show how view availability affects target evidence in these specific real UAV cases, rather than establishing general real-world multiview performance. Full details and numerical verification are given in Appendix A.2.6.

Chapter 4

Results

This chapter evaluates how controlled viewpoint diversity affects UAV object detection across training, evaluation, and inference-time multiview use. The detector benchmark first compares YOLOv8n, YOLOv8l, and Faster R-CNN across the five viewpoint-exposure regimes and identifies YOLOv8l-M4 as the strongest configuration. The subsequent fixed-detector analyses use this model to examine how performance varies across the azimuth-elevation-radius grid, showing that viewpoint sensitivity is systematic rather than random. Finally, the analysis extends beyond the controlled synthetic domain to assess whether these findings remain meaningful in real UAV imagery. Zero-shot evaluation is used to quantify the synthetic-to-real domain gap, after which YOLOv8l-M4 is fine-tuned on labelled real UAV images to test whether real-domain adaptation can recover detection performance. The resulting fine-tuned detector is subsequently used for same-object multiview and Shapley analyses on real UAV sequences, providing an assessment of whether the multiview patterns observed in the synthetic experiments also appear after adaptation to the real domain.

4.1 Experiment 1: Training-Side Viewpoint Diversity

4.1.1 Detector-Family Benchmark

The full M4 viewpoint-coverage regime produces the highest test-set point estimate within all three detector families, although performance across the intermediate restricted regimes is not monotonic (full fifteen-model comparison in Appendix A.3.1). YOLOv8l achieves the highest mAP50 : 95 within every training regime, improving from 0.4110 under M1 to 0.6396 under M4. YOLOv8l-M4 has the highest test-set point estimate in the evaluated benchmark and is therefore used as the fixed detector for the subsequent diagnostic analyses. This identifies the strongest evaluated detector-regime configuration, but should not be read as a controlled architectural comparison between one-stage and two-stage detection, since Faster R-CNN and YOLO differ in training schedule, input resizing, and augmentation (Appendix A.2.1).

Figure 4.1 condenses the benchmark to the selection metric used throughout the thesis. The markers show the test-set mAP50 : 95 point estimates, and the vertical bars show 95% class-bootstrap confidence intervals computed over the ten per-class AP50 : 95 values. The figure therefore shows both the training-regime effect and the detector-family effect, while also indicating the uncertainty due to class-to-class variation. All three detector families obtain their highest point estimate under the full M4 regime, and YOLOv8l-M4 gives the highest overall mAP50 : 95 point estimate. However, the overlap between the YOLOv8n

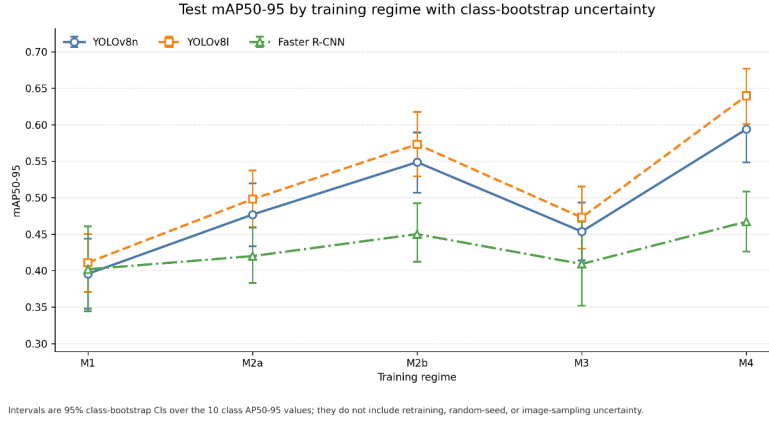


Figure 4.1: Test-set mAP50 : 95 by detector family and viewpoint-exposure regime. Markers denote the original mAP50 : 95 point estimates, and vertical bars denote 95% class-bootstrap confidence intervals over the tenper-class AP50 : 95 values. These intervals reflect class-to-class variability and do not include retraining or random-seed uncertainty. YOLOv8l-M4 has the highest point estimate in the benchmark.

and YOLOv8l intervals shows that their difference should be interpreted cautiously. The full multi-metric class-level results (Appendix A.3.1) show that the M4 advantage is broadly distributed across categories rather than driven by one or two classes.

4.1.2 Single-View Training

A complete 72-model sweep trains an individual YOLOv8l detector on each viewpoint cell alone and evaluates it on the full M4 test split, to quantify how restrictive single-view training is and how strongly this depends on which viewpoint is chosen. The mean single-view model reaches $\text{mAP50 : 95} = 0.3384 \pm 0.0046$, and the best (**elmid-radmid-az045**) reaches only 0.4164, far below the full M4 model’s 0.6396 while the worst (**ellow-radnear-az270**) reaches just 0.2678. Figure 4.2 shows this variation has regional structure across the viewpoint lattice rather than being isolated to one high-performing cell, indicating that no individual viewpoint provides sufficient appearance diversity to match broad-view training, even though viewpoint selection during training clearly matters.

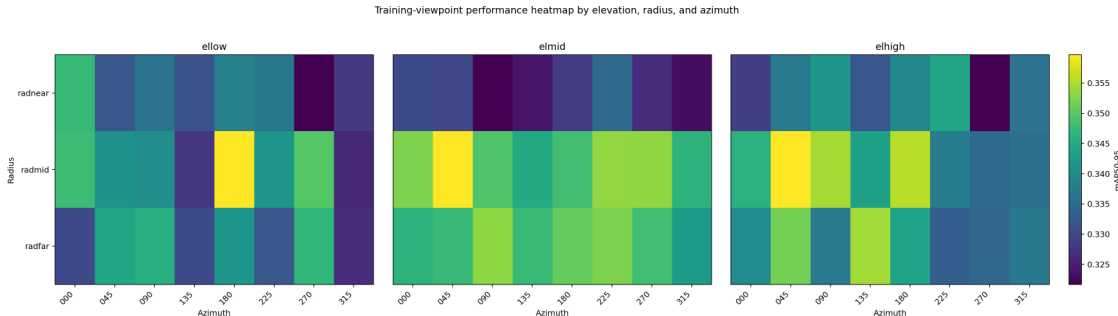


Figure 4.2: Test-set mAP50 : 95 obtained by training YOLOv8l on only one viewpoint at a time. The structured hot and cold regions show that training utility is not randomly distributed across the viewpoint lattice.

4.1.3 Observation Strength vs Training Strength

Viewpoint utility differs depending on whether a viewpoint is used as an inference-time observation for the full detector or as the sole training source for a restricted detector: across the 72 viewpoints, the Spearman rank correlation between the two is only 0.514, and none of the top-10 observation viewpoints appears among the top-10 training viewpoints (full single-view sweep and scatter plot in Appendix A.3.1). Deployment viewpoint selection and training-data selection are therefore separate design problems.

4.1.4 Pair-View Training

The pair-view sweep tests whether training on two viewpoints improves generalization over a single viewpoint, evaluating every restricted model on the same full M4 test split (Appendix A.3.1 shows the full single- vs. pair-view distribution). The best completed pair-view model reaches $\text{mAP}_{50 : 95} = 0.4958$, which is 0.0794 higher than the best single-view model. Across all completed models, the pair-view mean is also higher than the single-view mean: $\text{mAP}_{50 : 95} = 0.4207$ for pair-view training versus 0.3384 for single-view training.

This comparison is confounded by training-set size, since pair-view models use roughly twice as many images as single-view models. Table 4.1 therefore compares each restricted setting against an equal-budget control sampled broadly from the full M4 viewpoint space rather than from one or two fixed viewpoint cells.

Table 4.1: Restricted-view training compared against equal-budget controls sampled from the full M4 viewpoint space.

Setting	Train images	Validation images	mAP _{50 : 95}
Full M4 baseline	all	all	0.6396
Single-view mean	144	31	0.3384 [0.3294, 0.3475]
Equal-budget M4 (single-view mean)	144	31	0.4843
Single-view best	135	35	0.4164
Equal-budget M4 (best single)	135	35	0.4803
Pair-view mean	287	61	0.4207 [0.4185, 0.4228]
Equal-budget M4 (pair-view mean)	287	61	0.5234
Pair-view best	298	59	0.4958
Equal-budget M4 (best pair)	298	59	0.5196

At every matched budget, the equal-budget M4 control has a higher point estimate than the corresponding restricted setting. For example, the control matched to the pair-view mean reaches $\text{mAP}_{50 : 95} = 0.5234$ versus 0.4207, and the control matched to the best pair reaches 0.5196 versus 0.4958. Because the best and equal-budget control rows are single or selected point estimates in the available outputs, these differences are interpreted descriptively rather than as formal significance tests. The averaged restricted-view comparison is stronger: the pair-view mean exceeds the single-view mean under the configuration-bootstrap analysis. Overall, the fair-budget comparison suggests that the benefit of M4 comes from broader viewpoint coverage within a fixed image budget, not only from increasing the number of training viewpoints from one to two (Figure 4.3).

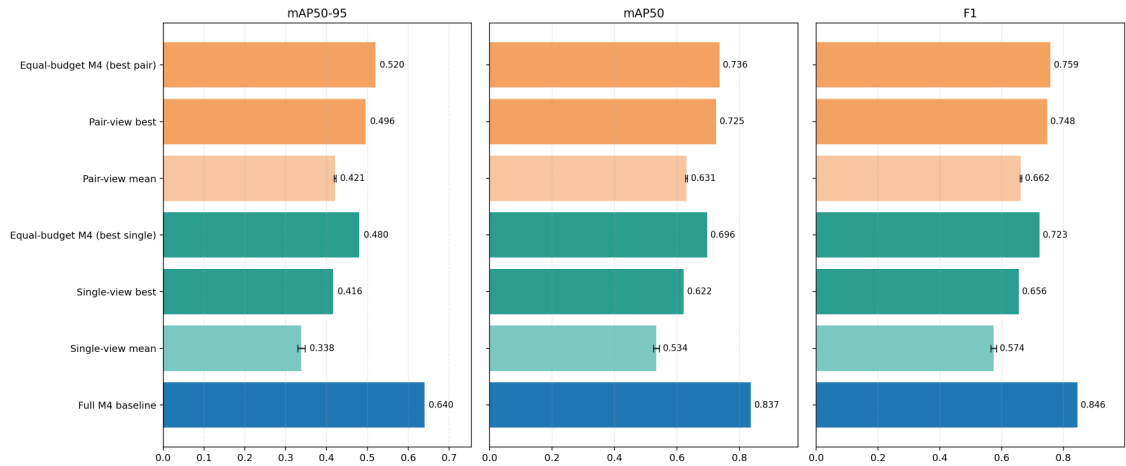


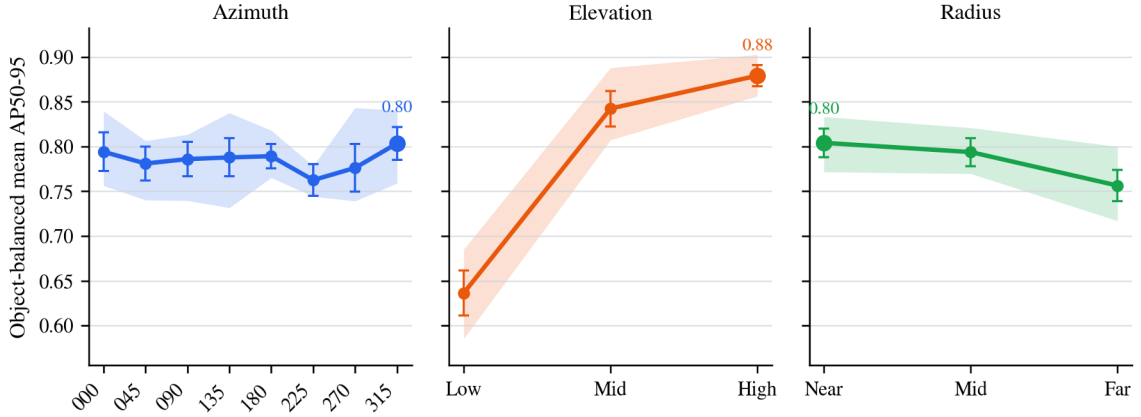
Figure 4.3: Fair training-side comparison against equal-budget M4 controls on the fixed M4 test split. Bars show mAP50 : 95, mAP50, and F1. Error bars are shown only for the single-view and pair-view mean rows, where they denote 95% configuration-bootstrap confidence intervals over completed trained viewpoint configurations. Unbarred rows are single or selected point estimates and are included as descriptive references.

4.2 Experiment 2: Fixed-Detector Viewpoint Sensitivity

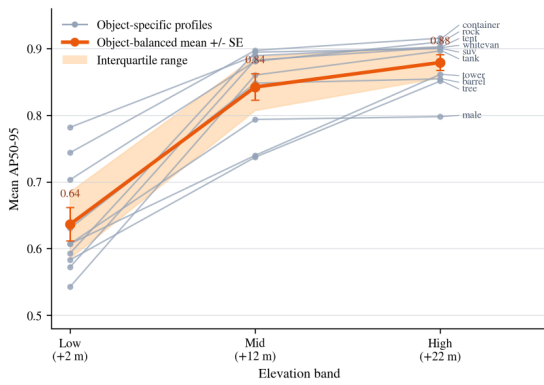
4.2.1 Factor-Level Viewpoint Dependence

With training exposure held fixed using YOLOv8l-M4, in the fixed-detector analysis, high elevation yields the highest mean detection-quality score for all ten classes (Figure 4.4b), while the object-balanced profile shows a substantially larger low-to-mid change than the subsequent mid-to-high change (Figure 4.4a). The object-balanced mean rises sharply from low to mid elevation and increases further at high elevation, with high elevation giving the highest mean score for all ten classes. The low-to-mid elevation difference is large relative to the associated standard errors, whereas the additional improvement from mid to high elevation is more moderate.

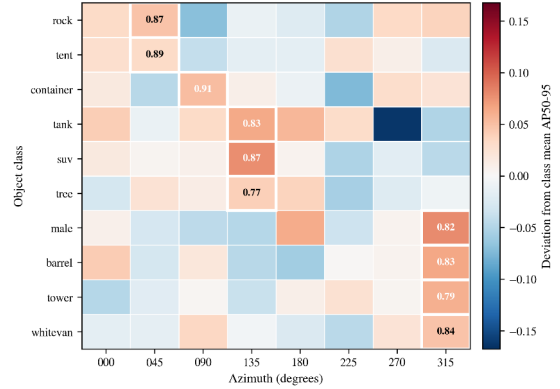
Azimuth shows a weaker and less stable effect. The object-balanced azimuth means are tightly clustered and their uncertainty ranges overlap more strongly than for elevation. Moreover, the azimuth profiles should not be interpreted as direct measurements of object-intrinsic symmetry. Each azimuth bin pools over elevation, radius, object instances, scale, occlusion, and scene context. Therefore, deviations from ideal bilateral or rotational symmetry can reflect dataset composition and detector bias rather than the physical symmetry of the objects themselves. Radius shows a smaller, predominantly scale-related effect (Figure 4.4). Thus, the factor-level analysis is used to compare the relative strength of viewpoint factors, not to infer a unique geometrically optimal view for each object class.



(a) Direct object-balanced mean AP50 : 95 profiles for azimuth, elevation, and radius. Elevation shows the steepest aggregate change, while azimuth and radius vary more modestly. The azimuth profile is a marginal performance trend, not a controlled object-symmetry measurement.



(b) Elevation profiles across classes. Thin grey lines are individual classes; the thick orange line is the object-balanced mean.



(c) Azimuth response heatmap after centering each class on its own mean. The class-specific structure should be interpreted cautiously because azimuth bins pool over elevation, radius, scale, and scene context.

Figure 4.4: Factor-level viewpoint dependence for YOLOv8l-M4. Panel a compares the direct aggregated level profiles for azimuth, elevation, and radius, panel b shows that class-level trajectories generally improve toward high elevation, and panel c shows class-specific azimuth variation. Because azimuth is analyzed as a marginal factor, these profiles should not be interpreted as controlled tests of bilateral or rotational object symmetry. Error bars and shaded regions indicate ± 1 SE for the aggregated profiles; the azimuth heatmap is centered within each class and is intended to show relative class-specific deviations rather than absolute performance.

As a sanity check on the azimuth profiles, we compared opposite azimuth pairs for approximately bilateral classes and tested azimuthal flatness for near-rotational classes. These checks did not show robust violations of the expected symmetries: no bilateral pair mismatch survived FDR correction, and barrel and tower did not significantly depart from flatness. This supports interpreting the azimuth trends as weak, marginal detector-performance patterns rather than as direct evidence of object-intrinsic viewing preferences (Appendix A.2.4). A supporting clutter-stratification check confirmed that detection quality decreases as scene clutter increases while binary detection remains largely unaffected, indicating that the benchmark retains meaningful difficulty variation (full

results in Appendix A.2.5).

4.3 Experiment 3: Inference-Time Multiview Analysis

4.3.1 Selection versus Evidence Accumulation

Best-box selection captures the rescue component of the multiview gain: mean scene-balanced strict quality increases from 0.873 ± 0.008 with one available view to 0.909 ± 0.006 with two views and 0.919 ± 0.005 with three. Thus, simply making another observation available improves the probability that at least one view contains a stronger target detection. The full method comparison, including threshold-rule and challenge-slice results, is provided in Appendix A.3.3. Evidence accumulation provides an additional improvement beyond this selection effect. Noisy-OR with best IoU reaches 0.951 ± 0.004 with two available views and 0.966 ± 0.003 with three, remaining the strongest evaluated fusion strategy alongside odds-product fusion with best IoU. The two-view Noisy-OR improvement over the single-view reference is substantially larger than the associated standard errors, supporting a meaningful accumulation benefit rather than a change in the mean that is small relative to the observed variability. Mean-IoU and support-weighted variants remain positive but lower, indicating that preserving the strongest available localization is more useful than averaging localization quality across observations. In both selection and accumulation families, the largest gain occurs when moving from one to two views, while the third view provides a smaller refinement.

4.3.2 Marginal Contribution Of Additional Views

To isolate how much the next available observation contributes on average, independent of any specific angle or position, Table 4.2 reports the permutation-based coalition-size marginal analysis. The largest gain occurs when moving from one to two available images. On the selection-style strict-quality game, the expected contribution of the second image is 0.036 ± 0.003 , compared with 0.010 ± 0.001 for the third image. The same pattern is stronger for the fusion-aware Noisy-OR + best-IoU game, where the second image contributes 0.079 ± 0.004 and the third contributes 0.015 ± 0.001 .

Table 4.2: Count-first image-count Shapley proxy using random within-scene coalition orderings. Values are reported as mean $\pm$ standard error (SE) across scene-level means.

Added image	Scene support	Best strict quality	Noisy-OR + best IoU	Marginal strict gain	Marginal Noisy-OR + best IoU gain
1	205	0.873 ± 0.008	0.873 ± 0.008	0.873 ± 0.008	0.873 ± 0.008
2	205	0.909 ± 0.006	0.951 ± 0.004	0.036 ± 0.003	0.079 ± 0.004
3	205	0.919 ± 0.005	0.966 ± 0.003	0.010 ± 0.001	0.015 ± 0.001
4	205	0.925 ± 0.005	0.971 ± 0.002	0.0055 ± 0.0004	0.0051 ± 0.0005
5	200	0.928 ± 0.004	0.974 ± 0.002	0.0037 ± 0.0003	0.0026 ± 0.0002
6	194	0.931 ± 0.004	0.976 ± 0.002	0.0028 ± 0.0002	0.0018 ± 0.0001
7	188	0.933 ± 0.004	0.977 ± 0.002	0.0022 ± 0.0002	0.0013 ± 0.0001
8	177	0.935 ± 0.004	0.978 ± 0.002	0.0018 ± 0.0002	0.0009 ± 0.0001

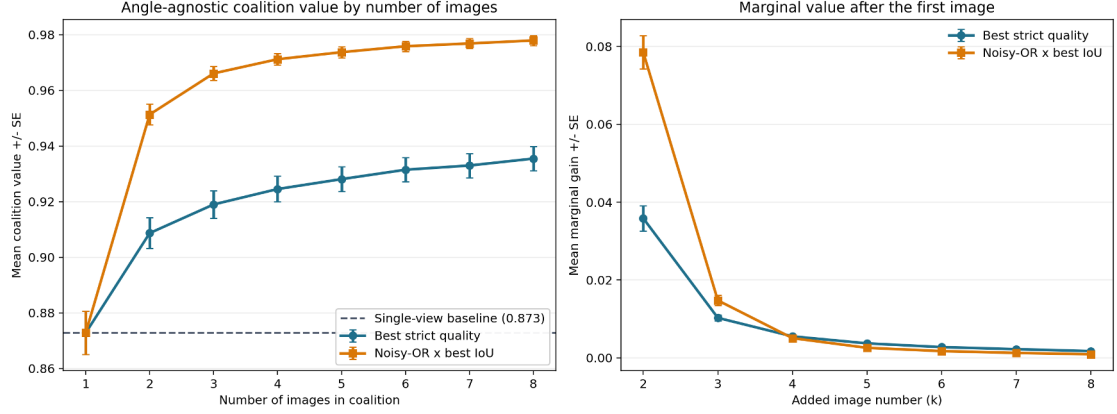


Figure 4.5: Permutation-based coalition-size marginal analysis with scene-level SE error bars. The left panel shows mean coalition value as the number of available images grows; the dashed horizontal line marks the single-view baseline. The right panel shows the expected marginal gain of each additional image after the first image. The largest gain occurs from one to two available views, while later gains are progressively smaller.

The decrease from second to the third observation is larger than the corresponding standard errors, supporting a diminishing-return pattern at the coalition-size level. Later observations remain positive in the mean, but their absolute size becomes progressively small: by the eighth available image, the marginal gains are only 0.0018 ± 0.0002 for selection and 0.0009 ± 0.0001 for Noisy-OR fusion. The main conclusion is therefore not that every additional view gives a clearly meaningful improvement, but that the largest operational benefit is concentrated in the first additional observation.

Scene support falls from 205 scenes at $k = 1-4$ to 177 at $k = 8$, since not every target instance has that many available views. Even so, the direction of the pattern remains stable: marginal gains shrink as coalition size increases. This result should be read as a coalition-size finding, not as an angle-ranking result or as evidence for a deployable end-to-end multi-UAV fusion system.

4.4 Experiment 4: Real-World Transfer and Adaptation

4.4.1 Zero-Shot Transfer and the Domain Gap

Zero-shot transfer is weak across all four real-world evaluation settings (Table 4.3; per-class breakdown in Appendix A.3.4). Performance on the internet-curated same-taxonomy dataset is somewhat higher than on VisDrone or AU-AIR, indicating that closer class alignment reduces one source of mismatch, but most labelled real objects are still missed everywhere, including on the self-collected UAV set, which retains the thesis’s own object taxonomy under real acquisition conditions. Controlled synthetic benefits therefore do not automatically translate into robust real-world detection.

Table 4.3: Aggregate zero-shot transfer performance across all four real-world evaluation settings. VisDrone and AU-AIR use approximate overlapping-class mappings to the synthetic taxonomy; the same-taxonomy dataset and self-collected UAV set use the ten synthetic classes directly. Full per-class results appear in Appendix A.3.4.

Dataset	Images	Instances	Precision	Recall	mAP ₅₀
VisDrone	277	9,525	0.0479	0.0114	0.0111
AU-AIR	32,823	127,265	0.0281	0.0136	0.0038
Same-taxonomy dataset	54	235	0.434	0.0555	0.0502
Self-collected UAV	156	592	0.1440	0.0501	0.0098

4.4.2 Self-Collected UAV Zero-Shot Evaluation

The self-collected UAV validation set is the clearest physical-world check in the thesis, using real drone imagery while retaining the thesis’s object taxonomy. Across 156 images and 592 labelled instances, the detector reaches $\text{mAP}_{50} = 0.0098$ and $\text{mAP}_{50 : 95} = 0.0044$, far below the synthetic M4 test performance of the same checkpoint ($\text{mAP}_{50} = 0.8367$, Appendix A.3.1). Only *suvs*, *whitevans*, *tanks*, and *containers* obtain non-zero mAP_{50} , and several classes are effectively unrecovered. Diagnostic plots (precision–recall curve and confusion matrix; Appendix A.3.4) confirm the dominant failure mode is missed detections rather than poor ranking. Most true objects are assigned to background, and the best all-class F1 is only about 0.02, so threshold tuning alone cannot rescue performance.

4.4.3 Real-Image Fine-Tuning on the Self-Collected UAV Dataset

To test whether the weak zero-shot performance reflects an adaptable synthetic-to-real gap, the YOLOv8l-M4 checkpoint was fine-tuned on the real-UAV training subset and compared with the original synthetic-only checkpoint on the same 39-image held-out test set. Across all ten classes, fine-tuning produces a large improvement. Precision rises from 0.015 to 0.638, recall from 0.007 to 0.514, F1 from 0.009 to 0.569, and mAP_{50} from 0.006 to 0.637, with true positives increasing from 1 to 74 and false negatives dropping from 143 to 70.

Because seven classes contain only one to three test instances, the primary comparison instead considers the three higher-support classes (*whitevan*, *suvs*, *male*; 130 of 144 labelled objects). For these classes, precision rises from 0.083 to 0.644, recall from 0.008 to 0.500, and mAP_{50} from 0.008 to 0.432 (Table 4.4). False positives increase from 11 to 36 in this subset, but alongside a large precision improvement, indicating the additional predictions are predominantly useful rather than indiscriminate; the effect of fine-tuning is therefore best characterized as improved target recovery and discrimination.

Table 4.4: Matched comparison of the synthetic-only and real-image fine-tuned YOLOv8l-M4 detectors on the same 39-image held-out real-UAV test subset. Higher-support classes are *whitevan*, *suv*, and *male*, each with at least 10 ground-truth instances. Precision, recall, and F1 are calculated at confidence ≥ 0.25 and IoU ≥ 0.50 .

Evaluation scope	Model	Precision	Recall	F1	mAP50	TP	FP	FN
All 10 classes	Synthetic-only	0.015	0.007	0.009	0.006	1	67	143
All 10 classes	Fine-tuned	0.638	0.514	0.569	0.637	74	42	70
Higher-support classes	Synthetic-only	0.083	0.008	0.014	0.008	1	11	129
Higher-support classes	Fine-tuned	0.644	0.500	0.563	0.432	65	36	65

At class level, AP50 for *whitevan* increases from 0.024 to 0.479, *suv* from 0.000 to 0.500, and *male* from 0.000 to 0.316, indicating the adaptation gain is not attributable to one dominant category. Several sparsely represented classes obtain high AP50 values after fine-tuning, but with only one to three ground-truth instances a small number of successful detections can produce apparently high scores, so these are treated as exploratory diagnostics rather than evidence of robust class-level generalization.

Taken together, these results show that the synthetic-only detector’s weak real-domain performance is not irreversible. Fine-tuning on 80 labelled real training images changes precision from 0.015 to 0.638, recall from 0.007 to 0.514, and mAP50 from 0.006 to 0.637 on the 39-image held-out test set. Because seven classes contain only one to three test instances, the higher-support subset provides the more reliable interpretation: for *whitevan*, *suv*, and *male*, mAP50 increases from 0.008 to 0.432. The experiment therefore demonstrates adaptation to this sampled real-UAV domain, not broad real-world deployment robustness.

4.4.4 Real Same-Object Coalition-Size Progression

The same-object case study contains five physical targets and 35 usable views. Coalition scores are first averaged within each physical object and subsequently macro-averaged across the five targets, so that objects with more available viewpoints do not receive greater weight. For the fine-tuned detector, binary detection is already high with one available view, reaching approximately 0.960 ± 0.041 , and increases to 0.996 ± 0.004 with two views and 1.000 ± 0.000 with three. The corresponding gains of $+0.036$ and $+0.004$ show a clear ceiling effect: most physical targets are already detected from a single observation, leaving relatively little room for additional views to improve binary target recovery.

Quality-based metrics retain considerably more headroom. Mean best strict target quality increases from approximately 0.696 ± 0.069 with one view to 0.778 ± 0.042 with two views and 0.812 ± 0.033 with three. The first additional view therefore produces a gain of $+0.082$, compared with $+0.034$ for the third view. The uncertainty intervals decrease as additional views become available, while the mean continues to increase, indicating that access to additional observations not only improves average target quality but also reduces some of the between-object variability in the available best observation. The heuristic Noisy-OR $\times$ best-IoU score shows the same diminishing-return pattern, increasing from approximately 0.696 ± 0.070 with one view to 0.859 ± 0.036 with two views and 0.912 ± 0.018 with three. Here again, the largest change occurs when moving from one to two available observations, while the third view provides a smaller additional improvement.

Within these five sequences, the main value of additional views after fine-tuning is therefore not converting large numbers of complete misses into detections, but increasing

the probability that at least one available image contains a stronger, better-localized target observation. Because the standard errors are calculated across only five physical-object-level means, these results should be interpreted as descriptive case-study evidence rather than precise population-level estimates. The synthetic-only checkpoint remains weak on the same real sequences. Its macro-averaged detection rate increases from approximately 0.080 ± 0.081 with one view to 0.133 ± 0.135 with two and 0.167 ± 0.168 with three. Mean strict target quality similarly increases only from approximately 0.046 ± 0.047 to 0.080 ± 0.081 and 0.105 ± 0.106 . In this case, the variability is of similar magnitude to the mean performance itself, so the small increases from additional observations should not be interpreted as strong evidence of a reliable multiview benefit. Multiple views cannot compensate for the underlying failure of the synthetic-only detector to recognize real-domain target appearances. Figure 4.6 summarizes these results and shows the macro-averaged means together with ± 1 SE across the five physical targets

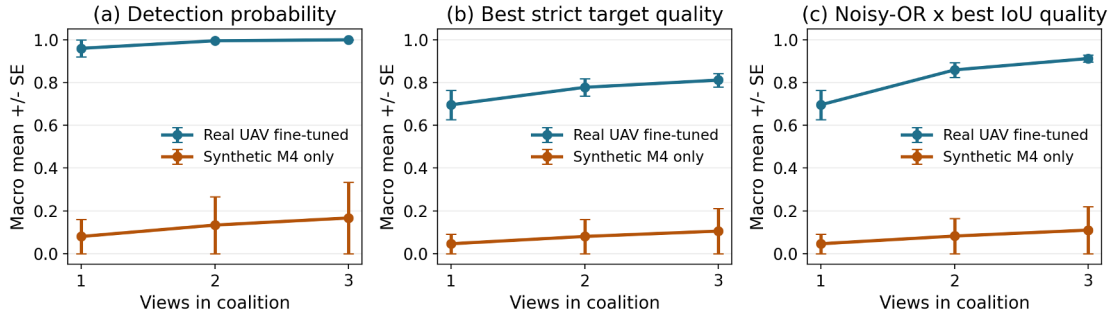


Figure 4.6: Macro-averaged performance as one, two, and three views of the same physical object become available: (a) detection probability, (b) best strict target quality, and (c) Noisy-OR $\times$ best-IoU quality. Coalition scores are first averaged within each physical object and subsequently macro-averaged across the five objects. The fine-tuned detector shows a ceiling effect in binary detection but continued improvement in target quality, with the largest gain occurring from one to two views. The synthetic-only checkpoint remains weak on the same real sequences. Error bars indicate variability across the five physical targets.

4.4.5 Exact Shapley Attribution Across Real Views

Exact Shapley values were computed over all usable observations within each physical-object sequence to determine how the available coalition value was distributed across individual views. The expected Shapley efficiency property was satisfied in every sequence: the individual view contributions summed to the corresponding grand-coalition value within numerical precision. In addition, the number of evaluated coalitions matched 2^n for each object, where n denotes the number of usable views. These checks confirm that the reported values correspond to exact, rather than sampled, Shapley attributions.

The resulting attribution patterns differ substantially between physical targets. For the parking-lot car, contributions range only from 0.118 to 0.158 across six observations, while the M truck similarly shows a relatively narrow range of 0.142–0.193 across five observations. In contrast, the Ooij tower exhibits a much larger spread of 0.035–0.237, and the La Souris truck ranges from essentially zero to 0.161. The white-truck sequence again shows a comparatively even distribution, with contributions between 0.112 and 0.158. Thus, within these real-world sequences, the multiview value is sometimes distributed

across several similarly useful observations, whereas for other targets it is concentrated in a smaller subset of particularly informative views.

Table 4.5 summarizes these object-level differences. Across the five fine-tuned sequences, the macro-averaged grand-coalition strict quality is 0.845 ± 0.025 , where the standard error is calculated across the five physical-target grand-coalition values. This object-balanced aggregation is consistent with the preceding real-world analysis. By comparison, the synthetic-only checkpoint achieves a substantially lower macro-average grand-coalition value of 0.169. Thirty of its 35 strict-quality Shapley contributions are near zero, compared with only one of the 35 contributions after real-image fine-tuning. These near-zero synthetic-only values should not be interpreted as evidence that the corresponding viewpoints are intrinsically uninformative; rather, they largely reflect the synthetic-only detector’s inability to recognize the targets reliably in the real domain.

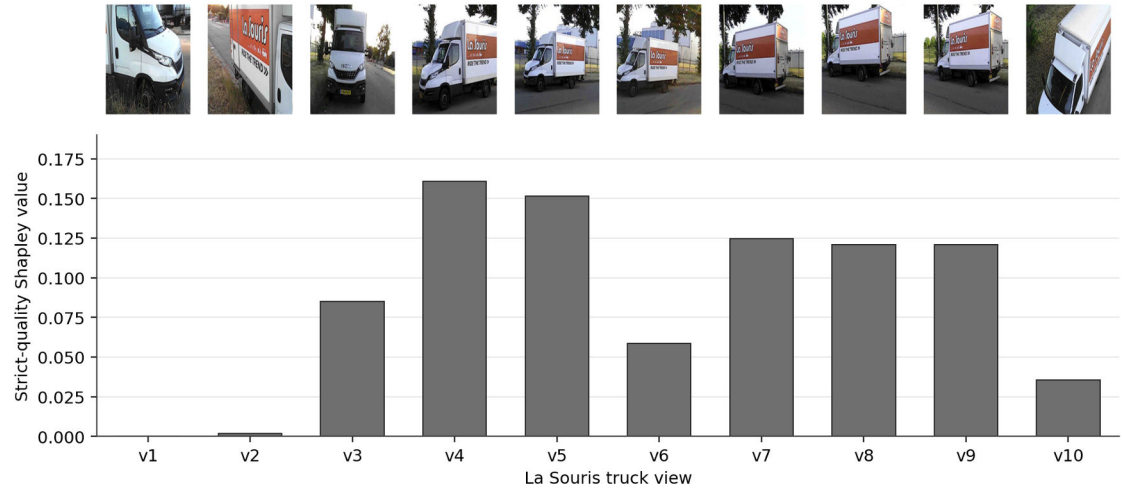
Table 4.5: Exact Shapley summary for the fine-tuned detector. Individual Shapley values sum to the corresponding grand-coalition strict-quality value. View labels v_i are object-specific indices within each real-image sequence and do not denote a shared azimuth, elevation, or camera distance across objects. Consequently, the best-view labels are interpretable only within each physical object and should not be compared geometrically across rows.

Object	Views	Grand-coalition quality	Shapley (min–max)	Best local view	Qualitative viewpoint
Parking car	6	0.789	0.118–0.158	v_1	elevated rear-right oblique
Ooij tower	7	0.801	0.035–0.237	v_7	elevated oblique
La Souris truck	10	0.860	0.000–0.161	v_4	front-left oblique
M truck	5	0.845	0.142–0.193	v_1	front oblique
White truck	7	0.929	0.112–0.158	v_1	elevated left side

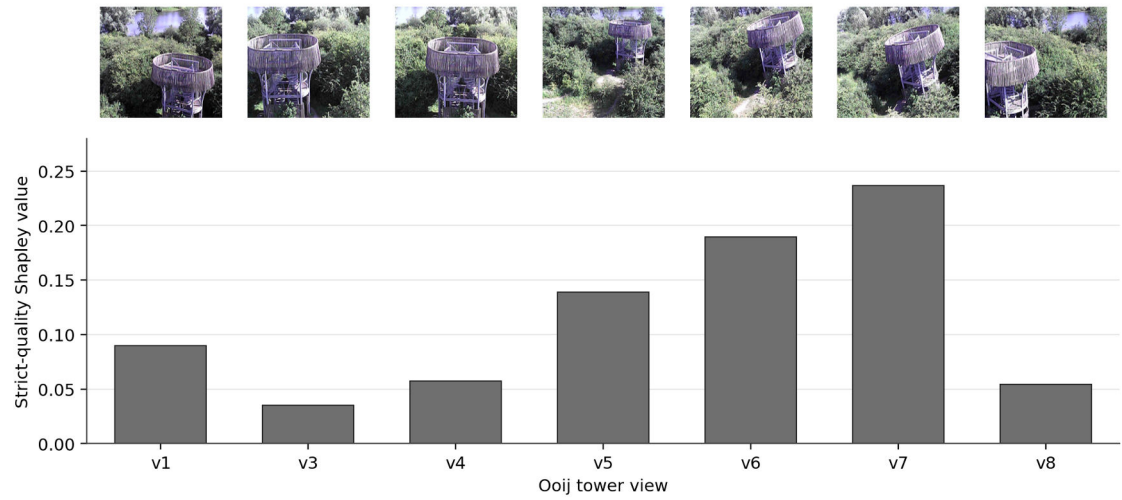
Single-view strict quality and exact Shapley contribution are strongly associated descriptively across the 35 observations ($r = 0.869$), although the 35 views are nested within only five physical targets and should therefore not be regarded as 35 independent samples. This relationship is expected from the coalition definition used here. Because the value of a coalition is determined by its strongest available observation, a view that already performs well in isolation can improve many coalitions in which it is introduced and therefore receives a larger Shapley contribution. The attribution should consequently be interpreted as quantifying each view’s marginal access to a strong observation, rather than as evidence of feature-level synergy between cameras.

Figure 4.7 relates these numerical attributions to the corresponding UAV observations. Because the real imagery was not acquired according to the controlled azimuth–elevation–radius grid used in the synthetic experiment, the labels $v_1, \dots, v_n$ are local to each physical-object sequence. For example, v_1 for the parking-lot car does not represent the same camera geometry as v_1 for the M truck. Accordingly, the analysis supports within-object statements about which observations are most valuable, but not a direct cross-object ranking of a shared geometric viewpoint.

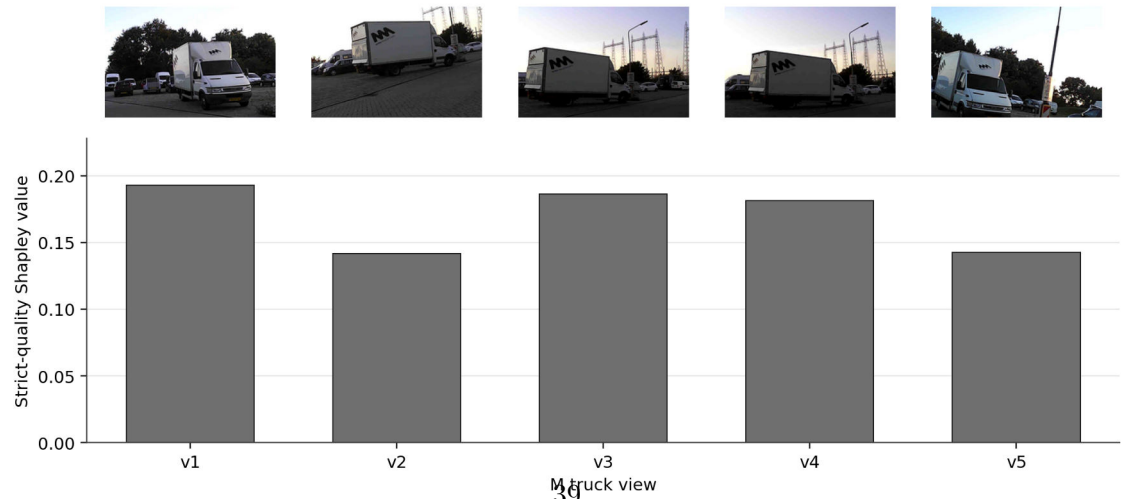
The qualitative patterns in Figure 4.7 reinforce the quantitative differences in Table 4.5. For the La Souris truck, the coalition value is concentrated particularly in views v_4 and v_5 , while the first two observations add almost no marginal value. The Ooij tower spans 0.035–0.237 in Shapley contribution, with v_7 receiving the maximum value. In contrast,



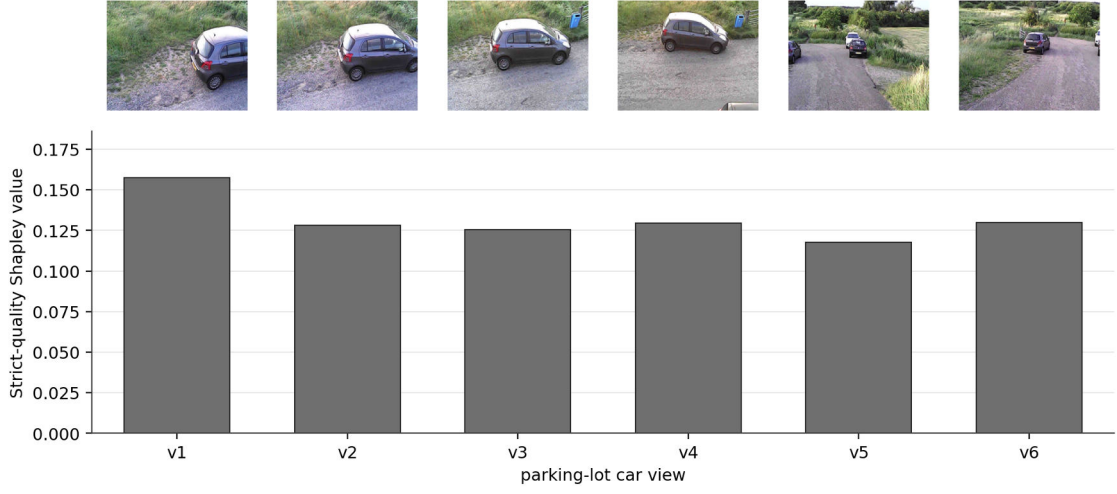
(a) La Souris truck. View importance is strongly non-uniform: views v4 and v5 provide the largest marginal contributions. The sequence spans 0.000–0.161, indicating that some observations add essentially no marginal strict-quality value while others account for a considerably larger fraction of the coalition value.



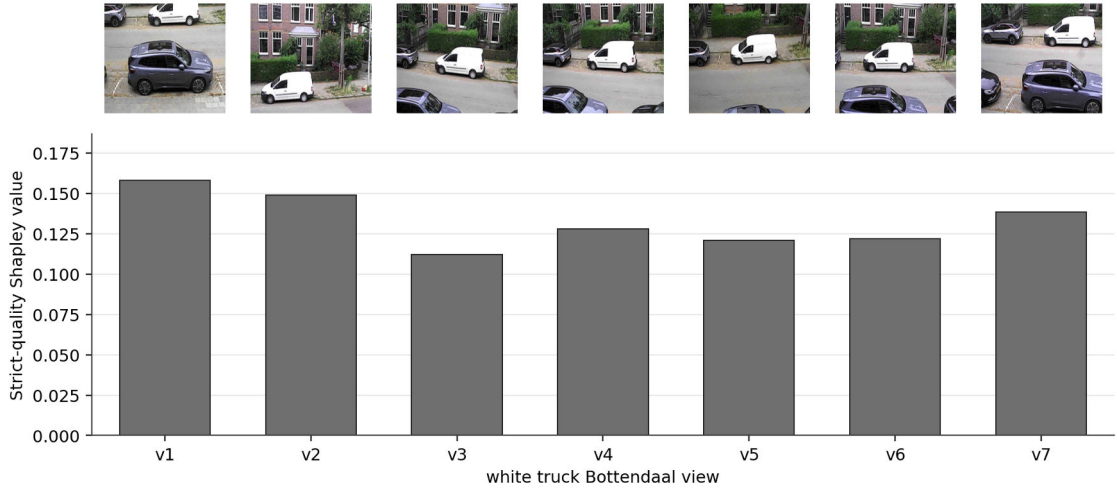
(b) Ooij tower. The Shapley contribution increases markedly for the later viewpoints, with v7 providing the largest contribution and v6 and v5 also being substantially more informative than the earlier observations.



(c) M truck. View importance is comparatively balanced across the sequence. Views v1, v3, and v4 make the largest contributions, while v2 and v5 remain useful but contribute somewhat less.



(d) Parking-lot car. Shapley values are distributed relatively evenly across all six observations, indicating little dependence on one uniquely informative viewpoint. Contributions span only 0.118–0.158 across the six parking-lot observations, indicating a comparatively flat attribution profile



(e) White truck. Contributions are again relatively distributed across viewpoints, although v1 and v2 are the strongest observations and v3 contributes the least.

Figure 4.7: Exact per-view strict-quality Shapley contributions for the fine-tuned real UAV model across five same-object sequences. Thumbnails show the observation associated with each contribution. The attribution patterns differ substantially between objects. For the La Souris truck and Ooij tower, performance is concentrated in a subset of particularly informative views, whereas the M truck, parking-lot car, and Bottendaal white-truck sequences show more evenly distributed contributions. This demonstrates that the value of an additional UAV observation is object- and viewpoint-dependent rather than being uniform across a sequence.

the M truck, parking-lot car, and white truck exhibit much flatter attribution profiles: no single observation accounts for a similarly dominant fraction of the available value.

The thumbnails provide a qualitative indication of why these patterns can arise. For the more uneven sequences, changes in camera position alter the visibility, apparent extent, and framing of the target sufficiently that some observations provide considerably stronger detections than others. For the flatter sequences, the target remains comparatively recognizable across a larger portion of the available viewpoints. These observations should remain descriptive, however, because the real sequences were not captured under controlled geometric conditions and therefore do not isolate azimuth, elevation, distance, background, scale, and occlusion independently.

Chapter 5

Discussion

This thesis examined viewpoint as three related but distinct components of UAV object detection: (i) the geometric coverage available during training, (ii) the quality of an individual observation after training, and (iii) the value of additional observations at inference time. Across these experiments, viewpoint mattered at each stage, but through different mechanisms. Broader training-viewpoint coverage improved generalization to held-out synthetic viewpoints; after training, individual observations remained systematically sensitive to camera-object geometry; and at inference time, additional views improved target-level performance primarily by increasing the likelihood of obtaining a stronger complementary observation, with further gains from evidence accumulation. The largest multiview improvement consistently occurs when moving from one to two available observations, after which the marginal benefit decreases.

The results also establish an important boundary to these controlled findings. Strong performance and systematic viewpoint effects within the synthetic benchmark do not imply robust transfer to real UAV imagery: the synthetic-only detector performs poorly across the real-world evaluation settings despite its broad geometric training coverage. Real-image fine-tuning substantially improves performance on the held-out UAV data, after which the same-object case study again shows a descriptive benefit from additional viewpoints. Taken together, these findings suggest that viewpoint diversity is valuable only within a sufficiently competent visual representation of the target domain. Geometric coverage can improve what a detector learns and additional viewpoints can improve what it observes, but neither compensates for a substantial mismatch between the training and deployment domains.

5.1 Experiment 1: Training-Side Viewpoint Diversity

The first question is what a detector learns when it is trained on narrow versus broad viewpoint exposure. The detector-family benchmark shows that models trained on a restricted set of viewpoints are less robust when evaluated on the full viewpoint distribution, while training with full geometric coverage across azimuth, elevation, and radius produces stronger detection performance on the standardized test split (Table A.8). The detector-family benchmark shows a consistent association between broader training-viewpoint coverage and higher test-set performance. For YOLOv8l, mAP50:95 increases from 0.4110 under M1 to 0.6396 under M4, while all three detector families obtain their highest point estimate under M4. The class-bootstrap intervals indicate that some architecture-level differences, particularly between YOLOv8n and YOLOv8l, should be interpreted cautiously. More importantly for RQ1, the equal-budget controls reduce the training-

set-size confound: at each matched image budget, broadly sampled M4 controls obtain higher point estimates than the corresponding restricted-view conditions. These results therefore support the interpretation that broader viewpoint coverage contributes beyond image count alone, although they do not isolate which specific viewpoint dimension or combination of viewpoints causes the improvement.

The improvement from restricted to broad viewpoint training is consistent with recent evidence that modern vision models are not inherently invariant to large perspective changes. In MVPOD, Jin et al. (2025) found pronounced cross-perspective degradation: for example, YOLOv5 trained on nadir-downward imagery obtained an AP of 0.689 when evaluated on the same perspective, but only 0.391 on oblique-downward imagery and 0.037 on horizontal imagery. Conversely, training on horizontal imagery produced an AP of 0.793 on horizontal test data but only 0.004 on nadir-downward imagery. MAVSD similarly reported improvements of up to 11% mIoU when training segmentation models with multiple UAV viewing angles rather than a vertical-only training set (Li et al., 2025). Although these studies use different datasets, tasks, and evaluation metrics and therefore cannot be compared numerically with the present results, they support the same broader conclusion: viewpoint coverage during training materially affects robustness outside the training perspective.

The present study extends this literature in two respects. First, the M1–M4 experiment samples viewpoint jointly across azimuth, elevation, and camera-to-object distance rather than treating perspective primarily as a discrete vertical domain. Second, the equal-budget controls separate geometric coverage from training-set size. The increase of YOLOv8l from 0.4110 under M1 to 0.6396 under M4 therefore cannot be interpreted only as another demonstration of cross-perspective degradation; the matched-budget experiments further indicate that distributing a fixed annotation budget across the viewpoint space is more useful than concentrating the same number of images in one or two viewpoint cells.

The intermediate training regimes show that not all forms of viewpoint diversity are equivalent, combining several azimuths with elevation variation at a fixed radius is more effective than varying only elevation or only azimuth. This suggests that useful diversity depends on how the viewpoint space is sampled, not only on whether more than one view is included. However, this comparison is confounded by training-set size. The main benchmark only demonstrates the practical advantage of broader exposure, while the equal-budget controls provide the stronger test of whether viewpoint diversity matters beyond image count alone.

The class-level pattern supports this interpretation. The largest gains from narrow to full viewpoint coverage appear for classes such as tank, container, suv, tree, and tent. For these categories, a single canonical view may provide a particularly incomplete representation of the class because different surfaces, shapes, and depth structures become visible as viewpoint changes.

5.1.1 Training Viewpoints and Observation Viewpoints

The distinction between training exposure and observation geometry is further supported by the comparison between strong training viewpoints and strong observation viewpoints. These two roles are related but not the same: a viewpoint favourable for immediate detection under the fully trained model is not necessarily the most useful viewpoint for training a detector that must generalize across the full viewpoint space.

A strong observation viewpoint is one from which the trained detector can detect the object reliably at test time. It may expose the target clearly, provide a favourable scale,

or reduce occlusion. A strong training viewpoint, by contrast, is useful if it helps the model learn representations that transfer to other viewpoints. A visually easy view is therefore not automatically the most informative training view; conversely, a view that is not individually optimal at test time may still be valuable during training because it exposes surfaces, foreshortening patterns, or occlusion states that would otherwise be missing. Here, an ‘easy’ viewpoint refers descriptively to an observation that yields high single-view detection quality under the fixed detector; it does not imply that the geometry is intrinsically easy independent of the detector or dataset.

This distinction has practical implications for data collection. If the goal is operational deployment, the relevant question is which viewpoint maximizes immediate detection reliability, whereas if the goal is training-data design, the relevant question is which viewpoints contribute most to transferable appearance coverage. Collecting more examples from the easiest-looking viewpoint may not be the most efficient strategy, a weaker but geometrically complementary view can be more useful for training than another repetition of an already well-represented one.

5.2 Experiment 2: Fixed-Detector Viewpoint Sensitivity

The training-side results establish that broader geometric coverage produces the strongest detector in the benchmark. The next question is what this detector reveals about viewpoint sensitivity itself, not what the model was shown during learning, but which viewpoints yield higher or lower detection quality at evaluation time once training exposure is held fixed.

5.2.1 Structured Viewpoint Dependence

In this fixed-detector analysis, elevation is the most consistent viewpoint factor: high elevation is the best-performing level for all ten object classes, indicating that vertical observation geometry has a broad influence on detection quality in the current benchmark. Two non-exclusive mechanisms could contribute to the high-elevation pattern. (i) Increasing camera height may reveal more upper-surface structure and reduce some forms of self-occlusion. (ii) Within the target-centered synthetic environment, higher views may also separate the projected target more clearly from nearby scene elements, reducing overlap and boundary ambiguity. The clutter-stratified analysis is consistent with the second mechanism because detection quality decreases as the number of competing labelled objects increases. However, elevation, target scale, and local framing were not manipulated independently, so the current experiment cannot determine how much of the observed elevation effect is attributable to either mechanism.

The more general point is that viewpoint dependence is *structured*: elevation shows a consistent cross-class direction, whereas azimuth’s optimum depends more strongly on object geometry (different classes have different informative sides as shape, surfaces, and silhouette change around the horizontal viewing circle), and radius is more closely tied to apparent object scale and class size. The benchmark therefore separates broad geometric trends from class-specific viewpoint preferences.

This result should not be over-interpreted as a universal rule that higher UAV viewpoints are always better. In real deployments, higher altitude can reduce the target’s pixel footprint, interact with sensor resolution, and introduce degradation from motion, atmosphere, or compression. Public UAV benchmarks repeatedly identify small-object scale as a main difficulty in aerial detection (Du et al., 2018; Tang et al., 2024; Hua and

Chen, 2025). The present benchmark can reveal a strong high-elevation advantage precisely because the target remains centered, visible, and within a controlled distance regime. The direction of the elevation effect is informative, but its magnitude is conditioned by the dataset design.

The direction of the elevation effect differs from some previous controlled UAV studies. Lee et al. (2022), for example, reported decreasing detector performance as the camera approached the nadir-view region. Their detectors, however, had not been trained systematically across those UAV viewpoints, and near-nadir observations differed substantially from the ground-level imagery represented in their original training distributions. In the present study, by contrast, the fixed YOLOv8l-M4 detector has explicitly been exposed to all three elevation bands during training, while camera-to-object distance and target visibility are bounded by the controlled capture design. The two results are therefore not necessarily contradictory. Rather, they suggest that observed viewpoint sensitivity depends jointly on the geometry of the observation and on the viewpoint distribution represented during training.

This interpretation is also compatible with MVPOD, where models perform strongly when training and test perspectives coincide but can deteriorate substantially under cross-perspective evaluation (Jin et al., 2025). The present experiment adds a finer-grained observation: even after broad-viewpoint training reduces this distribution mismatch, systematic geometric sensitivity remains. Thus, broad viewpoint exposure improves robustness but does not make the resulting detector viewpoint invariant.

This is part of the value of the controlled benchmark: by suppressing several sources of variation entangled in real UAV imagery, it can isolate a factor-level tendency that would otherwise be difficult to observe. The appropriate interpretation is not that the study identifies a universal optimal flight angle, but that vertical observation geometry should be treated as a primary design variable in dataset construction, evaluation, and multiview policy design.

5.2.2 Clutter and Benchmark Difficulty

The clutter-stratified analysis provides an interpretive check on dataset difficulty. Because the benchmark is target-centric and produces relatively high absolute scores, it could otherwise be read as too easy to differentiate meaningfully. The clutter results do not support that interpretation, since detection quality decreases as clutter increases even though the binary target-detected rate stays high throughout.

This clarifies the kind of difficulty that remains: clutter does not primarily determine whether the target is detected at all, but reduces the quality of the recovered detection by making localization, confidence ranking, or target separation less reliable. The target is usually still found, but recovery quality decreases as more competing objects are present, supporting the broader claim that, despite some ceiling compression from the target-centric setup, the dataset still contains systematic internal variation in difficulty.

5.3 Experiment 3: Inference-Time Multiview Gain

The final question is what happens when the detector is kept fixed and additional viewpoints become available only at inference time. This differs fundamentally from the training-side analysis: during training, viewpoint diversity determines the appearance variation from which the detector learns, whereas during inference it determines whether additional observations can compensate for limitations of a particular view. The multiview gain

observed here can be separated into two mechanisms. First, *rescue availability*: making an additional observation available increases the probability that at least one viewpoint provides a usable, well-localized detection. Best-box strict quality increases from 0.873 ± 0.008 with one view to 0.909 ± 0.006 with two. Second, *evidence accumulation*: combining matched evidence across observations provides an additional improvement beyond selecting the strongest single view, with Noisy-OR + best IoU reaching 0.951 ± 0.004 for two views. The experiment therefore provides evidence for both mechanisms, although the target-centric matched-box design does not establish how these gains would translate to scene-level multiview detection when target correspondence is unknown.

The transition from one to two available views is the most informative operational step, since the system can benefit from complementary geometry (differences in visibility, occlusion, scale, or orientation) that a single observation cannot provide. Park et al. (2023) similarly show that combining information across multiple camera views can improve detection when individual observations are incomplete. In the UAV setting, Nathan et al. (2023) demonstrate that coordinated observations can improve target visibility under occlusion. The present analysis adds a more mechanistic distinction by showing that part of the gain already arises from having a stronger alternative view available, while an additional improvement comes from accumulating matched evidence across views.

The threshold-family results show that rescue and confirmation represent different forms of multiview benefit. OR-based protocols improve availability-oriented metrics because they require only one usable view, while stricter confirmation protocols require support across multiple views and therefore optimize a more conservative criterion; their lower performance on availability metrics reflects that different decision rule, not a failure of multiview reasoning.

The challenge-slice results clarify where this fusion benefit is most visible: the gap between accumulation-based and selection-based fusion widens for weak-primary views, small target boxes, and hard scenes, suggesting cross-view accumulation is most useful when the primary observation is unreliable or incomplete. The larger gains on harder slices show the benchmark distinguishes between fusion mechanisms once the observation problem becomes more difficult, even though the global gain is bounded by a strong target-centric baseline.

These multiview results should be read as a perception-level contribution to future multi-UAV systems. They isolate the value of the observations such a system might acquire: before a swarm can decide where to move, it must first be possible to quantify which additional viewpoints improve target recovery and whether that improvement arises from rescue, confirmation, or evidence accumulation. Similarly, the fusion results are a target-centric matched-box analysis rather than full scene-wide multiview detection, camera calibration, cross-view identity tracking, and geometric reprojection are not available here, so the question answered is whether evidence accumulation improves target quality relative to selecting the best single matched box, once matched target evidence is already available.

The resulting interpretation is two-level: additional views help primarily by creating rescue availability, with the largest marginal step occurring between one and two available observations and returns decreasing thereafter. Once multiple associated detections are available, accumulation-based fusion consistently outperforms selection-based fusion as a family of methods.

The coalition-size analysis makes this diminishing-return pattern explicit. Under best-view selection, the expected contribution of the second observation is 0.036 ± 0.003 , falling to 0.010 ± 0.001 for the third. Under Noisy-OR evidence accumulation, the corresponding contributions are 0.079 ± 0.004 and 0.015 ± 0.001 . The reduction in marginal value is

therefore substantially larger than the uncertainty in the scene-level means. Additional observations beyond the third remain beneficial on average, but their absolute contribution becomes progressively smaller. This distinction is operationally important: an increase in the mean is not by itself sufficient to justify additional sensing, communication, or coordination resources if that increase is small relative to either its uncertainty or its practical cost.

This diminishing-return pattern complements multiview studies that commonly evaluate fixed camera counts or configurations, which can limit conclusions about how performance changes as sensing resources are added (Vora et al., 2023). The present analysis instead shows that the marginal value of an additional observation is largest for the second view and decreases thereafter.

5.4 Experiment 4: Real-World Transfer and Adaptation

The real-world experiments establish a boundary between internal validity within the controlled benchmark and external validity in real UAV imagery. Zero-shot transfer is consistently weak, demonstrating that systematic coverage of synthetic azimuth, elevation, and distance is not sufficient to overcome the broader appearance differences between simulation and reality. Real imagery introduces variation in texture, illumination, background, object instances, clutter, and image quality that is only weakly represented in the controlled synthetic environment. This domain gap may be amplified by the benchmark design, in which terrain, sky, and lighting are intentionally kept comparatively stable in order to isolate changes in camera-object geometry.

The fine-tuning experiment qualifies this result: limited real-domain supervision improves performance on the held-out real-UAV subset, showing the synthetic-trained detector remains adaptable once target-domain information is available. Rajpura et al. (2017) similarly show that synthetic imagery can provide a useful basis for object-detector training, while Tremblay et al. (2018) demonstrate that exposing a synthetic-trained model to real-domain data can help bridge the remaining reality gap. The present results are therefore consistent with the broader finding that synthetic and real data can serve complementary roles.

Importantly, the experiment does not isolate the causal contribution of synthetic pretraining to the final fine-tuned performance: the comparison is between a synthetic-trained detector before and after real-image fine-tuning. The results therefore support the conclusion that the synthetic-trained YOLOv8l-M4 detector is adaptable to the real domain, but do not establish that synthetic pretraining is more data-efficient or more accurate than real-only training.

This distinction reinforces rather than weakens the main contribution of the thesis. The synthetic benchmark remains valuable because it provides the experimental control required to isolate viewpoint-dependent mechanisms, while the real-world evaluation shows this internal control should not be confused with deployment realism. Synthetic and real data serve complementary roles: synthetic data supports controlled explanation of how viewpoint affects learning and inference, while real data is required to adapt and validate those mechanisms under the appearance variability encountered outside simulation.

This distinction is relevant to work such as RarePlanes, which explicitly compares synthetic and real training strategies for overhead object detection (Shermeyer et al., 2021). Because the present study does not include an equivalent real-only baseline, its contribution is evidence of adaptability after synthetic training rather than evidence that synthetic pretraining itself improves data efficiency.

5.4.1 Domain Competence Precedes Real Multiview Value

The real same-object analysis connects the synthetic multiview findings to the transfer experiment. For the synthetic-only checkpoint, increasing the number of available real views produces only small gains while absolute performance remains low. Multiple observations therefore do not compensate for a detector that does not adequately recognize the target appearance in the real domain. Fine-tuning provides the main transition from domain failure to useful real observations.

After fine-tuning, binary detection is already close to its ceiling from one image, while the multiview benefit becomes more visible in strict target quality. Across the five physical targets, mean strict target quality increases from 0.696 ± 0.069 with one available view to 0.778 ± 0.042 with two and 0.812 ± 0.033 with three. The corresponding marginal gains are therefore $+0.082$ and $+0.034$. Because uncertainty is estimated across only five physical targets, this pattern should be interpreted as descriptive case-study evidence rather than as a population-level estimate. The stronger evidence for diminishing multiview returns therefore comes from the synthetic scene-level analysis, while the real sequences show that the same qualitative mechanism can remain visible after domain adaptation.

This pattern provides a useful boundary to the multiview literature. Multiview detectors such as MVDET rely on complementary observations to recover information that may be weak or unavailable in one camera view (Park et al., 2023). At the acquisition level, coordinated UAV sensing can similarly improve target visibility when an individual viewpoint is affected by occlusion (Nathan et al., 2023). The present real-world case study suggests that such complementarity depends on sufficient domain competence: additional viewpoints do not compensate for a detector that poorly represents real target appearance, but become useful again after real-image fine-tuning. Domain adaptation and multiview observation therefore address different failure modes: adaptation determines whether the target can be recognized in the deployment domain, whereas multiview availability determines whether a stronger or more complementary observation can be obtained once that basic recognition capability exists. Given the five physical targets, this should be interpreted as a descriptive boundary condition rather than a general property of multi-UAV systems.

Chapter 6

Limitations and Future Work

6.1 Limitations

Split Structure and Generalization Boundary The most important methodological boundary concerns the dataset split. Although train, validation, and test are separated at the image and viewpoint-condition level, the same inferred synthetic object instances may occur across splits. Although train, validation, and test are separated at the image level, the same inferred synthetic object instances and viewpoint conditions may occur across splits. The dataset is therefore image-disjoint but not instance-disjoint. This is appropriate for isolating how viewpoint variation affects known objects under controlled pose changes, but does not establish generalization to entirely novel object instances, environments, or real-world scenes. The strongest claim supported by the benchmark is therefore that viewpoint diversity affects held-out-viewpoint performance for known synthetic object instances; future instance-disjoint evaluation is required to determine whether the same patterns generalize across object identity.

Synthetic and Controlled Dataset Design Simulation enables viewpoint to be manipulated systematically while many nuisance factors are held constant, but the current benchmark contains limited variation in background, weather, lighting, texture, target placement, and clutter. Keeping terrain, sky, and illumination comparatively stable is a deliberate internal-validity choice, because it prevents environmental appearance from becoming systematically associated with particular viewpoints. However, this also narrows the visual distribution encountered during training and may encourage reliance on scene-specific regularities. Viewpoint is therefore not tested under the full range of visual complexity found in real UAV imagery, and some clean factor-level effects, especially the strong elevation pattern, may partly depend on these controlled rendering conditions.

Target-Centric Framing and Visibility Assumptions The capture process places the target near the image center and ensures it is present in the frame. This is useful for isolating viewpoint effects on a known target, but reduces open-scene search difficulty, and creates a ceiling effect in some operational late-fusion metrics. The single-view target-found rate is already high, global differences among the strongest fusion methods are compressed. A related issue is the presence of target-absent and occluded cases: a small subset of test images contains no ground-truth instance of the filename target. These do not invalidate the detector-family benchmark, since standard evaluation uses the actual labels, but they complicate viewpoint-centric analyses that assume the filename target is visible, and

should be treated as part of the benchmark context rather than ordinary visible-target observations.

Controlled Diagnostic Multiview Analysis The late-fusion experiments operate on matched target observations rather than complete scene-level multiview detection: target correspondence is assumed, camera calibration and geometric reprojection are not modeled, and some diagnostic scores use localization information derived from evaluation. These assumptions make it possible to isolate rescue availability and evidence accumulation while keeping the detector fixed, but prevent the evaluated fusion rules from being interpreted as directly deployable algorithms. The multiview results should be read as controlled diagnostic analysis of the information contained in complementary observations, establishing whether additional associated views and corroborative evidence can improve target-level quality under controlled conditions. A deployable multi-UAV system would additionally require cross-view association, estimated camera geometry, communication between platforms, and a fusion mechanism operating without access to ground-truth-derived evaluation information.

Real-World Transfer Boundary The real-world transfer datasets do not provide controlled multi-angle observations of the same physical object, so none can evaluate target-level multiview rescue or fusion. The weak transfer results clarify the boundary of synthetic-only training, while the multiview conclusions remain supported by the controlled synthetic benchmark alone. VisDrone and AU-AIR further require approximate class mappings that do not fully match the synthetic taxonomy, while the internet-curated same-taxonomy dataset does not provide the scale or acquisition control of a dedicated UAV benchmark.

The real-image fine-tuning experiment adds a limitation of size and class imbalance: the held-out test split contains only 39 images and 144 objects, 130 of which belong to just three classes (*suv*, *whitevan*, *male*), so high AP values for the remaining sparsely represented classes are unstable and are not read as robust class-level generalization. The experiment evaluates a narrow real-image distribution, so improvement on the held-out subset demonstrates adaptation to the sampled UAV domain rather than robustness across platforms, cameras, locations, weather, altitudes, or acquisition protocols.

Limited Real Same-Object Coverage. The self-collected data contain five physical targets with repeated observations, enabling an exploratory real multiview analysis. However, the sequences are not controlled (360°) sweeps and do not independently vary azimuth, elevation, distance, scale, lighting, occlusion, or background. They therefore cannot validate the specific geometric effects identified in the synthetic benchmark.

Small Number of Independent Targets. Although 35 views and 2,752 model-object coalitions were evaluated, the independent experimental units were only five physical targets. Three were vans or trucks, creating additional class imbalance. Coalition enumeration improves within-object attribution but does not increase the number of independent samples.

6.2 Future Work

Increasing Dataset Realism and Intra-Class Variability Object classes should contain greater intra-class variability. Instead of representing each class with only a limited

number of object models, future datasets should include multiple instances per class that differ in shape, size, color, texture, and fine-grained structure. The same principle should be extended to the environment. Rather than keeping terrain, background, lighting, and object placement largely fixed, future versions should render each azimuth–elevation–radius condition across multiple independently varied scene configurations.

Importantly, environmental factors should be crossed with viewpoint rather than changed systematically together with it. For example, the same viewpoint could be rendered under several terrain layouts, lighting conditions, backgrounds, and object placements. This would preserve the ability to estimate the effect of camera–object geometry while testing whether the observed viewpoint dependencies remain stable across more realistic scene variation. Such a factorial design would improve external validity without reintroducing the viewpoint–scene confounding that the controlled benchmark was designed to avoid. This direction is related to domain-randomization approaches, which deliberately increase synthetic appearance variation to reduce reliance on simulation-specific visual regularities (Tobin et al., 2017).

Stricter Generalization Splits The present benchmark is appropriate for held-out-viewpoint analysis on known object instances. If the goal is to evaluate deployment robustness, future datasets should include instance-disjoint and preferably scene-disjoint splits, so that no object instance or environment layout appears in both training and testing.

Adaptive Viewpoint Planning Future work should move from fixed viewpoint selection toward adaptive viewpoint planning. The present results show that the second viewpoint provides the largest marginal gain and that complementary geometry is more valuable than simply repeating an individually strong view. This naturally motivates an active-perception setting in which the system decides where a second UAV or camera should move after observing the first image. Coordinated UAV sensing has already shown that viewpoint selection can improve target visibility under occlusion (Nathan et al., 2023). The present findings suggest a related question: whether the viewpoint-sensitivity and coalition-value estimates identified here could be used to decide where that second observation should be acquired. This would be especially relevant for weak-primary cases, where the current analysis shows the clearest multiview benefit.

Stronger Multiview Geometry Future work could incorporate calibrated camera poses, epipolar constraints, explicit cross-view association, or learned multiview feature fusion. Methods such as MVDET demonstrate how information from multiple camera views can be combined within a dedicated multiview detection framework (Park et al., 2023), while cross-view constraints have also been used to support association and detection across cameras (Nassar et al., 2019). Incorporating such mechanisms would move the present analysis beyond ground-truth-anchored matched boxes and toward a more realistic collaborative perception system. It would also test whether the complementarity observed in the current two-view results can be converted into stronger localization, better false-positive suppression, and more reliable scene-level detection.

Scene-Wide Multiview Object Detection The current thesis asks how multiple views help recover a known target under controlled viewpoint variation. A richer operational benchmark would instead ask how multiple UAVs jointly detect all relevant objects in a cluttered scene, where target identity and correspondence are not known in advance. This

would bring the experimental setting closer to multiview detection systems that operate on complete scenes rather than matched target observations (Park et al., 2023). Such an extension would test whether the rescue and evidence-accumulation mechanisms identified here remain visible once association errors, false positives, and competing objects become part of the problem.

Evaluation Protocol Refinement Global averages remain useful, but they should be reported together with challenge-slice results, such as weak-primary views, small target projections, low-IoU observations, and high-clutter scenes. This would make it easier to distinguish methods that perform similarly on high quality, target-centered observations but differ under more difficult conditions. In bounded and ceiling-compressed metrics, it may also be useful to report how much of the remaining headroom to the metric ceiling is recovered by a stronger method.

Value of Synthetic Pretraining for Real-Data Efficiency A direct extension of the fine-tuning experiment is to isolate whether synthetic pretraining reduces the amount of real data required for adaptation. This requires a matched comparison between at least two initialization strategies: a detector trained under the synthetic M4 regime and subsequently fine-tuned on real imagery, and the same detector architecture initialized from the same generic pretrained weights but fine-tuned directly on the identical real-image subset. Evaluating both models on the same held-out test data would separate the contribution of synthetic pretraining from the contribution of real-image supervision itself.

This comparison would connect the present adaptability result to work that more directly evaluates the value of synthetic data relative to real supervision. For example, Tremblay et al. (2018) investigate synthetic-to-real training strategies under limited real-domain supervision, while RarePlanes explicitly compares synthetic and real training data for overhead object detection (Shermeyer et al., 2021). Repeating the comparison with progressively larger real training subsets would reveal whether synthetic pretraining provides its greatest benefit in the low-data regime and whether any advantage persists as more real annotations become available.

Sequence-Disjoint Real Multiview Evaluation Future data collection should include more physical objects per class, keep all views of a physical target in the same data split, and reserve complete target sequences for final evaluation. Standardized pose metadata would allow real-world coalition value to be related back to azimuth, elevation, and distance without assuming a full 360-degree capture for every target.

Chapter 7

Conclusion

This thesis investigated how controlled viewpoint diversity affects UAV object detection at three distinct stages: (i) how geometric coverage during training relates to held-out-viewpoint performance, (ii) how a fixed detector remains sensitive to camera-object geometry after training, and (iii) how additional observations affect target-level performance at inference time. A fourth experiment evaluated zero-shot real-world transfer and real-image adaptation, establishing the external-validity boundary of the controlled synthetic analyses.

Regarding RQ1, broader training-viewpoint coverage is associated with higher held-out-viewpoint performance for known synthetic object instances. Across all three detector families, M4 yields the highest point estimate within each family. The equal-budget controls further support a contribution of viewpoint coverage beyond image count: at every matched budget examined, the broadly sampled M4 control obtains a higher point estimate than the corresponding restricted condition. Because several of these control comparisons are based on single trained configurations rather than repeated optimization runs, they should be interpreted as supporting evidence rather than formal significance tests. These conclusions are restricted to the present image-disjoint, non-instance-disjoint synthetic benchmark and do not establish generalization to unseen object instances.

Regarding RQ2, fixed-detector performance remains systematically dependent on viewpoint even after broad viewpoint training. Elevation produces the largest and most directionally consistent cross-class variation, whereas azimuth shows weaker and more class-dependent variation and radius has a smaller, predominantly scale-related role. The supplementary azimuth-symmetry analysis provides no robust evidence that the observed marginal azimuth profiles violate expected object symmetries after correction for multiple comparisons. The elevation result is therefore best interpreted as a benchmark-specific geometric effect, rather than evidence that higher UAV viewpoints are universally optimal in real-world operation.

Regarding RQ3, the clearest inference-time multiview benefit occurs when moving from one to two available observations. Additional views provide value through two mechanisms: rescue availability, where at least one view yields a usable target detection, and evidence accumulation, where matched detections from multiple views strengthen the target-quality score. On the strict target-quality metric, the two-view accumulation methods improve beyond best-view selection, indicating that multiview value is not only a rescue effect. However, the differences among the strongest accumulation methods, and many of the later two-to-three-view gains, are small relative to the reported uncertainty and should therefore be interpreted cautiously. The most defensible conclusion is a family-level one: preserving the best localization while accumulating confidence evidence is more effective

than pure selection, mean-IoU fusion, or strict confirmation in this target-centric diagnostic. These findings do not constitute evidence for the end-to-end performance of a deployable multi-UAV fusion system, because cross-view association, geometric reprojection, camera calibration, communication, and autonomous viewpoint planning were not modeled. The exploratory real same-object analysis showed a qualitatively similar pattern after real-image adaptation: across five physical targets, the one-to-two-view transition produced the clearest improvement in target quality, while the third view gave only a smaller mean increase. Because binary detection was already near its ceiling, the main benefit was not a large increase in detection probability, but a higher chance of obtaining a strong target observation.

The exploratory real-image Shapley analysis further showed that view importance was object-dependent. For some physical targets, a small subset of observations accounted for much of the available coalition value, whereas other targets showed more evenly distributed contributions across views. Because the real sequences were not acquired on a shared controlled pose grid, these results support within-object attribution of informative observations rather than a common cross-object ranking of geometric viewpoints.

Regarding RQ4, zero-shot transfer of YOLOv8l-M4 is weak across all four real-world evaluation settings: mAP50 ranges from 0.0038 on AU-AIR to 0.0502 on the same-taxonomy real-image dataset, and reaches 0.0098 on the self-collected UAV set. Fine-tuning on 80 labelled real training images increases performance on the separate 39-image held-out real-UAV test set. For the three classes with at least ten test instances, mAP50 increases from 0.008 to 0.432, precision from 0.083 to 0.644, and recall from 0.008 to 0.500. Because the test set is small and class-imbalanced, these results support adaptation to the sampled real-UAV domain rather than broad deployment robustness. Moreover, because no matched real-only baseline was evaluated, the improvement cannot be attributed specifically to synthetic pretraining; the experiment establishes adaptability of the synthetic-trained checkpoint, not superiority over direct real-image fine-tuning.

Together, the experiments support a more bounded conclusion: (i) controlled synthetic data provides a useful instrument for isolating viewpoint-dependent mechanisms, (ii) the synthetic-trained checkpoint does not transfer reliably to the evaluated real UAV domains without adaptation, and (iii) after real-image fine-tuning, the five-target case study shows descriptive additional quality gains from multiview availability. These findings do not establish a universally optimal UAV viewpoint configuration or prove that synthetic pretraining is superior to real-only training, but they identify where viewpoint diversity contributes within the controlled pipeline and where domain mismatch limits those benefits.

At the same time, controlled viewpoint analysis addresses only one dimension of UAV object detection. The weak zero-shot transfer results demonstrate that systematic azimuth-elevation-distance coverage cannot replace the texture, sensor, environmental, and intra-class variability of natural UAV imagery. The central contribution of this thesis is therefore not a deployment-ready detector or multiview UAV system, but a controlled experimental framework for disentangling how viewpoint affects detector learning, observation sensitivity, and the informational value of additional views. Future work can build on this framework through greater asset and scene diversity, instance-disjoint evaluation, multi-seed training, real-domain adaptation, and multiview methods that perform association and fusion without ground-truth assistance.

Appendix A

Appendix

A.1 Background

Table A.1: Representative datasets relevant to viewpoint-aware UAV perception.

Dataset	Main characteristics	Viewpoint treatment	Viewpoint limitation
Archangel (Shen et al., 2023)	Hybrid real/synthetic human-detection benchmark with UAV-position and object-pose metadata.	Includes position and pose metadata for analyzing UAV-object geometry.	Human detection only; no repeated azimuth–elevation–distance sampling across classes.
AU-AIR (Bozcan and Kayacan, 2020)	Traffic-monitoring dataset covering multiple altitudes and outdoor conditions.	Camera tilts provide vertical viewpoint diversity.	No systematic azimuth or distance sampling around fixed objects.
Lee et al. (Lee et al., 2022)	Game-engine-generated UAV imagery for controlled detector validation.	Varies camera position, viewing angle, illumination, and object pose.	No dense object-centric viewpoint grid or separate training, sensitivity, and inference analyses.
MAVREC (Dutta et al., 2023)	Multiview dataset combining aerial and ground perspectives.	Synchronized observations across platforms.	Platform-centric; views are not systematically sampled around fixed targets.
MAVSD (Li et al., 2025)	Controlled multi-angle aerial dataset for segmentation.	Controls camera pitch using discrete viewing angles.	Limited to pitch variation; no full azimuth–elevation–distance control.
MDOT (Zhu et al., 2021b)	Multi-drone dataset of shared targets in dynamic scenes.	Multiple drones observe the same targets from different viewpoints.	Viewpoint is not systematically controlled around fixed targets.
MVPOD (Jin et al., 2025)	Benchmark for cross-platform vertical viewpoint shifts.	Models vertical perspective as a distribution shift.	Focuses on domain shift rather than repeated object-centric sampling.
SynDrone (Rizzoli et al., 2023)	Large synthetic dataset with RGB, depth, and point-cloud outputs.	Provides broad camera-angle and environmental variation.	Not designed for dense repeated sampling per object instance.
UAVDT (Du et al., 2018)	Real-world traffic dataset with altitude, camera-view, and environment labels.	Provides coarse viewpoint annotations along flight trajectories.	Viewpoint remains entangled with scene, identity, occlusion, and altitude.
UEMM-Air (Yao et al., 2025)	Multimodal synthetic dataset across urban and highway scenes.	Includes multiple heights and pitch angles.	Emphasizes modality and scene diversity rather than isolated viewpoint effects.
VisDrone (Zhu et al., 2021a)	Large real-world UAV benchmark with diverse scenes and object scales.	Viewpoint varies naturally through altitude, angle, and motion.	No controlled object-centric sampling around the same target.

A.2 Methods

This appendix collects implementation-level detail that supports the reproducibility of Chapter 3 but is not required to follow the main experimental argument.

A.2.1 General Methodological Detail

Exact Training-Regime Composition

Table A.2: Exact training subsets used throughout the detector benchmark.

Regime	Cells	Images	Training subset
M1	1	138	Only the canonical cell <code>elmid-radmid-az000</code> (0° azimuth, mid elevation, mid radius).
M2a	3	422	Elevation-only variation at fixed azimuth and radius: <code>elallow-radmid-az000</code> , <code>elmid-radmid-az000</code> , and <code>elhigh-radmid-az000</code> .
M2b	12	1714	Four cardinal azimuths (0°, 90°, 180°, 270°) crossed with low/mid/high elevation on the mid-radius ring.
M3	8	1144	All eight azimuths (0° to 315° in 45° steps) on the fixed <code>elmid-radmid</code> ring.
M4	72	10332	Full multiview training over all azimuth, elevation, and radius combinations.

M2a, M2b, and M3 provide intermediate controls over elevation and azimuthal coverage between the single-view (M1) and full-grid (M4) extremes. All regimes are evaluated on the same full-viewpoint validation and test splits.

Camera-Object Geometry: Coordinate Parameterization

The viewpoint grid described in Section 3.2.2 is defined by azimuth θ , distance r , and an elevation band implemented as a fixed vertical offset h above the target:

$$x_c = x_t + r \cos(\theta), \quad (\text{A.1})$$

$$y_c = y_t + r \sin(\theta), \quad (\text{A.2})$$

$$z_c = z_t + h. \quad (\text{A.3})$$

Annotation Format and Target-Box Selection

The dataset is automatically annotated from rendered segmentation outputs. Each YOLO label row has the form (c, x_c, y_c, w, h) , where c is the class index and $x_c, y_c, w, h \in [0, 1]$ are the normalized box center, width, and height. For an image of width W and height H , the pixel-space corners are

$$x_1 = W \left(x_c - \frac{w}{2} \right), \quad y_1 = H \left(y_c - \frac{h}{2} \right), \quad x_2 = W \left(x_c + \frac{w}{2} \right), \quad y_2 = H \left(y_c + \frac{h}{2} \right).$$

The target-centric analyses in Chapter 3 require identifying the intended focus object within images that may contain multiple boxes. Each capture has a manifest record containing the file name, the intended focus object, the camera pose, the target world

coordinates, and the azimuth–elevation–radius cell. When several boxes are present, the focus-object box is recovered by combining the manifest with the YOLO labels. Let $c^\star$ denote the focus-object class inferred from the manifest and let $\mathcal{B}_{c^\star}$ be the set of boxes of that class in the image. The target box is selected as

$$b^\star = \arg \min_{b \in \mathcal{B}_{c^\star}} \sqrt{(x_c(b) - 0.5)^2 + (y_c(b) - 0.5)^2},$$

that is, the box of the focus class closest to the image center.

Full Dataset Audit

Table A.3 reports the audited size and split structure of the synthetic dataset referenced in Section 3.3. Table A.4 reports the corresponding per-class image counts across splits.

Table A.3: Audited structure of the synthetic dataset.

Quantity	Value
Total images	14,760
Train images	10,332
Validation images	2,214
Test images	2,214
Inferred unique object instances	205

Table A.4: Synthetic dataset composition by object class.

Class	Total	Train	Val	Test	Instances
tent	1296	910	181	205	18
tank	1440	1023	217	200	20
tower	936	610	161	165	13
container	1584	1117	230	237	22
whitevan	1368	963	224	181	19
suv	1584	1098	245	241	22
male	1440	1028	200	212	20
rock	1512	1047	230	235	21
barrel	1440	1021	207	212	20
tree	2160	1515	319	326	30

Training Hyperparameters

Table A.5 reports the full archived training configuration for each detector family referenced in Section 3.4. Settings are held fixed within each detector family across all five viewpoint-exposure regimes (M1–M4); the only intended experimental variable is the training-viewpoint subset. Some implementation differences remain *between* detector families because YOLOv8 and Faster R-CNN use different frameworks, preprocessing policies, training schedules, and default assumptions.

Table A.5: Training protocol details.

Setting	Value
Validation / test protocol	Common held-out <code>val</code> and <code>test</code> splits for all detector families and regimes. Faster R-CNN training evaluates on the shared validation split every 500 iterations.
Input resize policy	YOLOv8n and YOLOv8l use 640 px inputs. Faster R-CNN inherits Detectron2 shortest-edge resizing with training-side choice from 640, 672, 704, 736, 768, or 800 px, together with a maximum training size of 1333 px and a test-time shortest edge of 800 px with maximum size 1333 px.
Training duration	YOLOv8n: 120 epochs. YOLOv8l: 100 epochs. Faster R-CNN: 1500 iterations.
Batch size	YOLOv8n: 16. YOLOv8l: 16. Faster R-CNN: <code>IMS_PER_BATCH=2</code> .
Initialization	YOLOv8n: pretrained <code>yolov8n.pt</code> . YOLOv8l: pretrained <code>yolov8l.pt</code> . Faster R-CNN: COCO-pretrained Detectron2 checkpoint for the Faster R-CNN R50-FPN 3x model-zoo configuration.
Optimizer / key settings	YOLOv8n and YOLOv8l use archived Ultralytics settings with <code>optimizer=auto</code> , <code>lr0=0.01</code> , <code>lrf=0.01</code> , momentum 0.937, and weight decay 5×10^{-4} . Faster R-CNN uses Detectron2 SGD defaults with momentum 0.9 and weight decay 10^{-4} , while the archived script sets the base learning rate to 2.5×10^{-4} , maximum iterations to 1500, disables warmup, and clears step decays, yielding a constant learning rate across training.
Augmentation / preprocessing	YOLOv8n: translation 0.1, scale 0.5, horizontal flip 0.5, mosaic 1.0, mixup 0, copy-paste 0. YOLOv8l: translation 0.1, scale 0.5, horizontal flip 0.5, mosaic 0, mixup 0, copy-paste 0. Faster R-CNN follows Detectron2 BGR preprocessing with pixel mean [103.53, 116.28, 123.675], pixel std [1.0, 1.0, 1.0], default horizontal flipping, and no enabled crop augmentation.
Random seed / determinism	YOLOv8n: seed 0, deterministic enabled. YOLOv8l: seed 0, deterministic enabled. Faster R-CNN: the archived training script does not set a fixed seed, so Detectron2 default <code>SEED=-1</code> .
Hardware	YOLOv8n and YOLOv8l were archived as GPU runs. The recovered Faster R-CNN launcher requests GPU runs.
Experimental control	Within each detector family, settings were kept fixed across M1–M4; the intended experimental variable is only the training-viewpoint subset.

Standardized Evaluation Metrics

All trained detectors are evaluated on the same test split using one standardized pipeline, so every model is scored under the same matching and aggregation rules. The reported metrics are precision, recall, F1, AP50, and mAP50 : 95:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad \text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}, \quad (\text{A.4})$$

where TP , FP , FN are true/false positives and false negatives after one-to-one class-wise box matching, so that missed detections, duplicate detections, and localization errors all affect the reported scores.

Average precision and $\text{mAP}_{50:95}$ are defined in Section 3.5; in particular, the primary model-ranking metric follows Equation 3.1. The appendix provides the remaining implementation details of the standardized matching and evaluation procedure.

A.2.2 Experiment 1: Training-Side Viewpoint Diversity

Single-View Training Sweep

To quantify viewpoint-specific training utility independently of the regime comparison, a complete 72-model sweep trains one YOLOv8l detector per individual viewpoint cell (using only that cell’s M4 training images) and evaluates each on the full M4 test split.

This also allows the two roles of a viewpoint, as an inference-time observation under the full detector, versus useful as the sole training condition for a restricted-view detector, to be compared directly, consistent with work treating view selection as an explicit design problem (Hou et al., 2024).

Pair-View Training Sweep

To test whether training on two viewpoints improves generalization beyond either alone, a detector is trained on the combined images of every unordered pair of viewpoint cells ($\binom{72}{2} = 2,556$ pairs) and evaluated on the same full-viewpoint test split. Each pair-trained detector is compared against the *better* of its two single-view baselines, asking whether adding a second specific training viewpoint improves performance beyond simply choosing the stronger of the two alone.

A.2.3 Experiment 2: Fixed-Detector Viewpoint Sensitivity

A.2.4 Azimuth Symmetry Sanity Check

The factor-level azimuth analysis in Experiment 2 is a marginal detector-performance analysis rather than a controlled physical symmetry experiment. Nevertheless, a sanity check was performed to test whether the observed azimuth profiles strongly contradict simple symmetry expectations. For approximately bilateral classes (**tank**, **whitevan**, **suv**, and **male**), opposite azimuth bins were compared: $000^\circ\text{--}180^\circ$, $045^\circ\text{--}225^\circ$, $090^\circ\text{--}270^\circ$, and $135^\circ\text{--}315^\circ$. For near-rotational classes (**barrel** and **tower**), azimuthal flatness was tested across all eight azimuth bins.

Figure A.1 shows the bilateral-class azimuth profiles. The whitevan class is closest to the expected opposite-view similarity, with a maximum opposite-pair difference of 0.052 AP50 : 95. The largest nominal mismatches occur for tank at $090^\circ\text{--}270^\circ$ and male at $135^\circ\text{--}315^\circ$, but no opposite-pair mismatch remains significant after FDR correction. Figure A.2 shows the near-rotational classes. Barrel and tower show visible variation across azimuth, but neither class shows a statistically significant departure from flatness.

These results indicate that the marginal azimuth profiles do not provide strong evidence against the expected object symmetries, but they also do not recover those symmetries cleanly. This is expected because each azimuth bin pools over elevation, radius, object instances, scale, occlusion, and scene context. The symmetry check therefore supports interpreting azimuth as a weak and confounded marginal detector-performance factor, rather than as a direct measurement of object-intrinsic viewing symmetry.

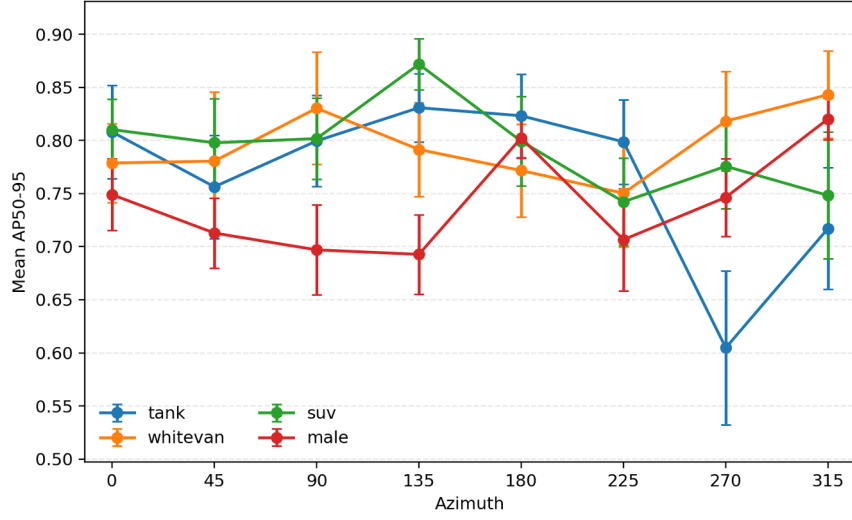


Figure A.1: Azimuth profiles for approximately bilateral object classes. Points show mean AP50 : 95 per azimuth bin and error bars indicate standard errors. Opposite azimuth pairs were compared as a symmetry sanity check, but no pairwise mismatch survived FDR correction.

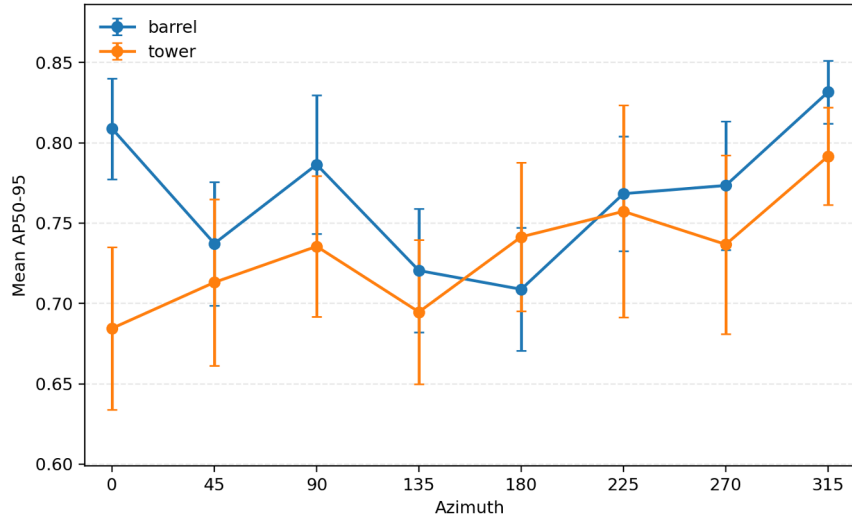


Figure A.2: Azimuth profiles for near-rotational object classes. Points show mean AP50 : 95 per azimuth bin and error bars indicate standard errors. Barrel and tower show no significant departure from azimuthal flatness in the aggregate test.

Table A.6: Summary of the azimuth symmetry sanity check.

Class	Test	Largest deviation	FDR-significant result
tank	Opposite-pair comparison	0.195 at 090°–270°	No
whitevan	Opposite-pair comparison	0.052 at 135°–315°	No
suv	Opposite-pair comparison	0.124 at 135°–315°	No
male	Opposite-pair comparison	0.128 at 135°–315°	No
barrel	Azimuthal flatness	range = 0.123, $q = 0.630$	No
tower	Azimuthal flatness	range = 0.107, $q = 0.870$	No

A.2.5 Experiment 3: Inference-Time Multiview Analysis

Threshold Family

For a single view u , let $c(u)$ be the matched target confidence at IoU50 and $i(u)$ the matched IoU. The strict target-quality score is $q(u) = c(u) i(u)$, an evaluation-time diagnostic (since $i(u)$ requires the ground-truth box) rather than a score available to a deployed detector. Threshold rules based on ranked per-view scores characterize how evidence changes as more observations become available. If S is a set of available views with ordered scores $q_{(1)}(S) \geq \dots \geq q_{(|S|)}(S)$, the threshold family is $v_r(S) = q_{(r)}(S)$, where r is the number of observations required to agree: $r=1$ gives the 1-of- k ("at least one") rule, $r=2$ gives 2-of-3 confirmation, and $r=3$ gives 3-of-3 unanimous support. For $r=1$, $v(S) = \max_{u \in S} q(u)$; the strongest available observation determines the coalition score. The same logic is applied to the per-image quality proxy and to binary target-found indicators.

Additional Fusion Variants

Additional reference variants (mean quality, unanimous best box, Noisy-OR + mean IoU, odds-product corroboration) are used to interpret family-level patterns. For matched confidences c_i and IoUs i_i in a coalition, noisy-OR confidence pooling is

$$c_{\text{fuse}} = 1 - \prod_i (1 - c_i), \quad q_{\text{fuse}} = c_{\text{fuse}} \max_i i_i \quad (\text{Noisy-OR} + \text{best IoU}),$$

with Noisy-OR + mean IoU replacing $\max_i i_i$ by the coalition mean, and support-weighted noisy-OR further multiplying by the support ratio $m/|S|$ (the fraction of views with matched support). The odds-product variant replaces confidence pooling with

$$o_{\text{fuse}} = \prod_i \frac{c_i}{1 - c_i}, \quad c_{\text{fuse}} = \frac{o_{\text{fuse}}}{1 + o_{\text{fuse}}},$$

coupled to the mean matched IoU.

Fusion-Aware Coalition Value (Synthetic Multiview)

The thesis uses a permutation-based coalition-size marginal analysis to quantify how the marginal value of an observation changes as more images become available, without attributing value to specific angles, positions, or UAV agents. Each available image in a scene is a player; its Shapley value averages its marginal contribution to coalition value $v(S)$ over all possible coalition contexts:

$$\phi_i = \sum_{S \subseteq I_s \setminus \{i\}} \frac{|S|!(|I_s| - |S| - 1)!}{|I_s|!} (v(S \cup \{i\}) - v(S)).$$

To isolate how value grows as the coalition grows from one view to two, two to three, and so on, the repository samples 256 random orderings π_s of each scene’s available images and tracks the coalition-growth sequence $S_k(\pi_s) = \{\pi_s(1), \dots, \pi_s(k)\}$, using two coalition values: a selection rule $v_{\text{best}}(S_k) = \max_{u \in S_k} q(u)$, and a fusion rule $v_{\text{fuse}}(S_k) = (1 - \prod_{u \in S_k} (1 - c(u))) \max_{u \in S_k} i(u)$ (Noisy-OR confidence pooled with the best available matched IoU). The marginal contribution of the k th added image is $\Delta_k(\pi_s) = v(S_k(\pi_s)) - v(S_{k-1}(\pi_s))$, with $S_0 = \emptyset$; scene-level means are averaged over permutations and then over all scenes with at least k available images. This produces an angle-agnostic proxy for the expected value of the 2nd, 3rd, 4th, and later view, without restricting to one viewpoint family or ranking individual azimuths.

Challenge Slices and Metric Roles

The ordered-pair fusion cache is re-evaluated on harder slices (weak primary quality, small target box, low primary IoU, far primary view, hard scenes, and primary-miss-only rows) to test whether the gap between accumulation-based and selection-based fusion widens when the primary observation is harder. Metrics are used asymmetrically by design: binary target-found rate is the most direct availability metric; target AP50 : 95 applies to threshold protocols, which still reduce to choosing one of several discrete matched detections; and target strict quality is the only score well-defined for every coalition family (a bounded 0–1 confidence-localization product), and is therefore used as the headline metric whenever all methods are compared in one ranking.

Clutter Stratification: Definition and Supporting Results

This analysis does not address RQ2 directly, since clutter is not a component of camera-object viewpoint; it instead checks whether the target-centric benchmark retains meaningful variation in detection difficulty. Clutter is operationalized from the label inventory as $\kappa_i = n_i^{\text{label}} - n_i^{\text{target}}$, the number of non-target labeled objects visible in image i , with a scene-level mean $\bar{\kappa}_s = \frac{1}{|V_s|} \sum_{i \in V_s} \kappa_i$ over the set of evaluated views V_s . Images are grouped into fixed bins (0, 1–4, 5–9, 10+) for descriptive summaries.

As a validity check on the target-centric benchmark design (Section 4), scenes were grouped into clutter quartiles by their mean count of non-target labeled objects. Table A.7 summarizes the resulting groups: detection quality declines monotonically with clutter, while the binary target-detected rate remains high throughout, indicating that clutter degrades localization and confidence quality rather than determining whether the target is found at all.

Table A.7: Scene-level clutter quartiles and mean target-centric performance under YOLOv8l-M4.

Group	Scenes	Mean clutter	Mean target localization-quality proxy	Mean strict quality	Mean detected rate
Q1 low clutter	53	3.2828	0.8927	0.9052	0.9917
Q2 mid-low clutter	52	4.7796	0.8675	0.8792	0.9772
Q3 mid-high clutter	49	6.0879	0.8402	0.8691	0.9762
Q4 high clutter	51	9.4917	0.8038	0.8375	0.9787

A.2.6 Experiment 4: Real-World Transfer and Adaptation

Exact Shapley Analysis (Real Same-Object Case Study)

Predictions were obtained using both the synthetic YOLOv8l-M4 checkpoint and the real-image fine-tuned checkpoint. Duplicate filenames were removed, and one Ooij image was excluded because no tower ground-truth box was available. A secondary Noisy-OR $\times$ best-IoU score was retained as a heuristic sensitivity measure rather than interpreted as a calibrated probability.

All within-object coalitions of one, two, and three views were evaluated. Results were first averaged across coalitions of equal size within each object and then macro-averaged across the five objects, preventing sequences with more viewpoints from receiving greater weight. Exact Shapley values were subsequently computed separately for each object and model by enumerating all $2^{|N|}$ subsets of the available view set N , with $v(\emptyset) = 0$. The primary coalition value was the best strict quality, $v_{\text{strict}}(S) = \max_{i \in S} q_i$, while binary detection rescue, $v_{\text{det}}(S) = \max_{i \in S} d_i$, and the heuristic Noisy-OR $\times$ best-IoU score were evaluated as secondary value functions. For each view i , the exact Shapley value was calculated as

$$\phi_i = \sum_{S \subseteq I_s \setminus \{i\}} \frac{|S|!(|I_s| - |S| - 1)!}{|I_s|!} (v(S \cup \{i\}) - v(S)).$$

Across both checkpoints, 2,752 model-object coalitions were evaluated. Correctness was verified by requiring the number of subsets to equal $2^{|N|}$ and the Shapley efficiency condition $|\sum_{i \in N} \phi_i - v(N)| < 10^{-8}$.

A.3 Results

This section collects supplementary results supporting Chapter 4, organized by the same four experiments.

A.3.1 Experiment 1: Training-Side Viewpoint Diversity

Full Detector-Family Benchmark

Table A.8 reports precision, recall, F1, AP50, and mAP50 : 95 for all fifteen detector-regime combinations referenced in Section 4. Figure A.4 shows the same comparison across all five metrics, confirming that the M4 advantage for YOLOv8l holds consistently across detection and localization measures rather than being an artifact of any single metric.

Table A.8: Standardized test-set performance across detector families and training regimes. Within each regime, the best mAP50 : 95 score is highlighted in bold.

Regime	Detector	Precision	Recall	F1	mAP50	mAP50 : 95
M1	YOLOv8n	0.8182	0.5372	0.6486	0.6013	0.3956
	YOLOv8l	0.8248	0.5588	0.6662	0.6280	0.4110
	Faster R-CNN	0.8476	0.5473	0.6651	0.6343	0.4019
M2a	YOLOv8n	0.8887	0.6408	0.7447	0.7023	0.4768
	YOLOv8l	0.8914	0.6711	0.7657	0.7254	0.4979
	Faster R-CNN	0.8605	0.6070	0.7119	0.7015	0.4199
M2b	YOLOv8n	0.9330	0.6937	0.7957	0.7608	0.5486
	YOLOv8l	0.9295	0.7389	0.8233	0.7922	0.5731
	Faster R-CNN	0.8712	0.6364	0.7355	0.7270	0.4501
M3	YOLOv8n	0.8220	0.5767	0.6778	0.6440	0.4536
	YOLOv8l	0.8331	0.6085	0.7033	0.6641	0.4728
	Faster R-CNN	0.8568	0.5800	0.6917	0.6690	0.4091
M4	YOLOv8n	0.9462	0.7355	0.8276	0.7978	0.5938
	YOLOv8l	0.9487	0.7880	0.8609	0.8367	0.6396
	Faster R-CNN	0.8919	0.6340	0.7412	0.7350	0.4671

Figure A.3 reports the per-class breakdown underlying the aggregate detector-family comparison, showing that the M4 advantage for YOLOv8l is not concentrated in a small number of classes.

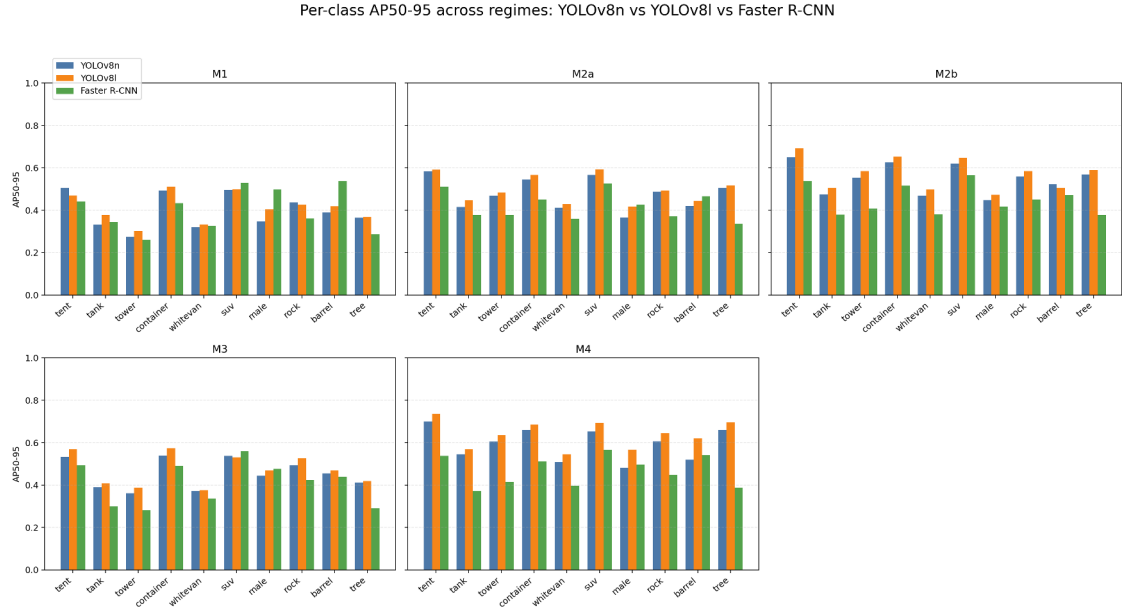


Figure A.3: Per-class AP50:95 across detector families and training regimes.

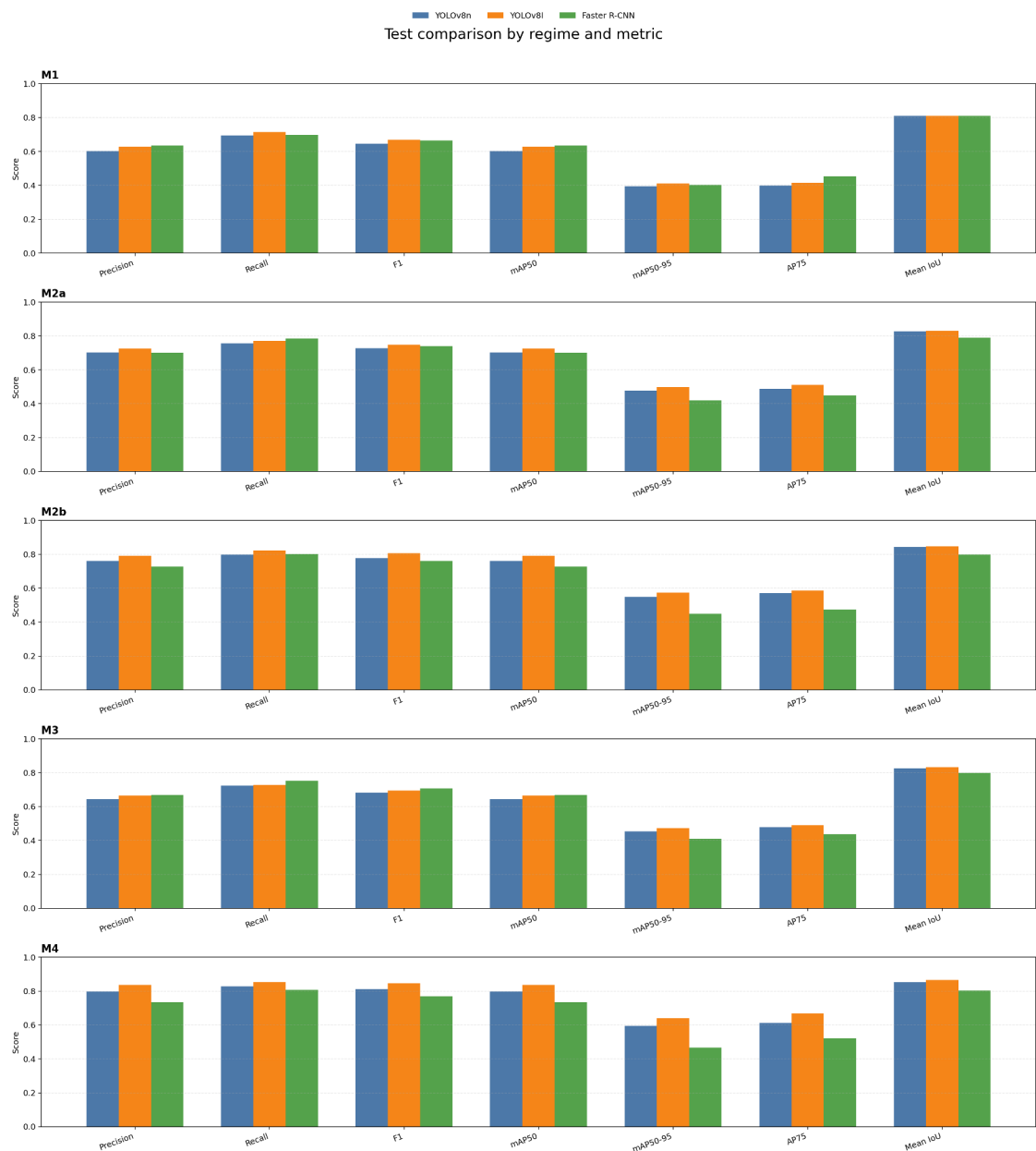


Figure A.4: Detector-family comparison across all five viewpoint-exposure regimes.

Pair-View vs. Single-View Distribution

Figure A.5 shows the full distribution of single-view and pair-view training performance referenced in Section 4, underlying the summary statistics reported there.

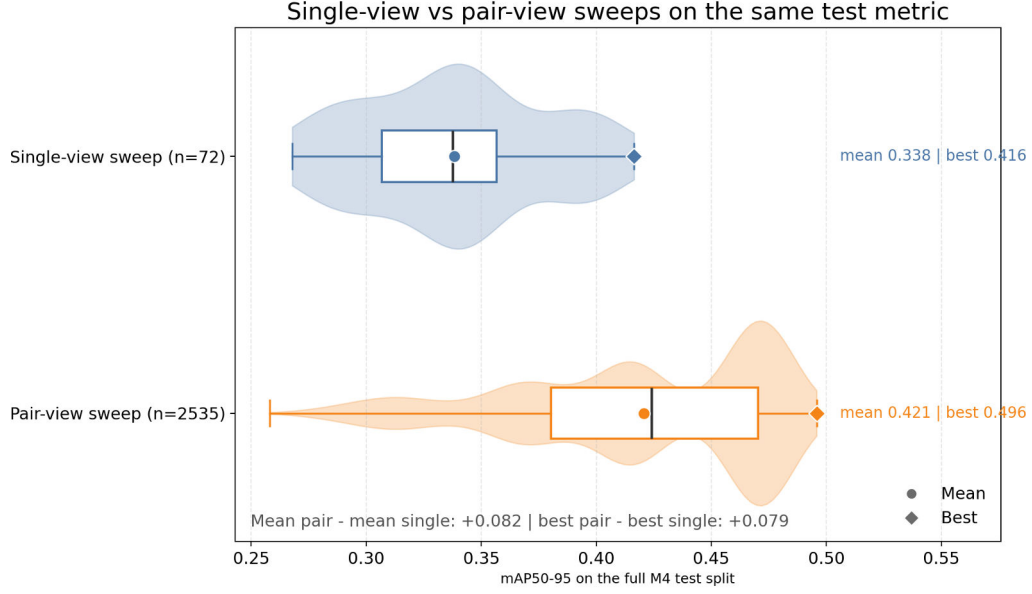


Figure A.5: Single-view and pair-view training sweep comparison on the full M4 test split. The violin and box plots show the distribution of $\text{mAP}_{50:95}$ across all restricted single-view models ($n = 72$) and all completed pair-view models ($n = 2556$).

Observation Strength vs. Training Strength

Viewpoint utility is evaluated in two distinct roles: *observation strength* (how well the fully trained M4 detector performs when the target is viewed from a given viewpoint) versus *training strength* (how well a detector trained only on that viewpoint generalizes to the full test grid). Across the 72 viewpoints, the Spearman rank correlation between the two is only 0.514, and none of the top-10 observation viewpoints appears among the top-10 training viewpoints. The best observation viewpoint is `elhigh-radnear-az225`, while the best training viewpoint is `elmid-radmid-az045`. Figure A.6 shows the resulting dispersed scatter, confirming that the most useful viewpoint at inference time is not necessarily the most useful viewpoint for training a restricted model; deployment viewpoint selection and training-data selection are separate design problems.

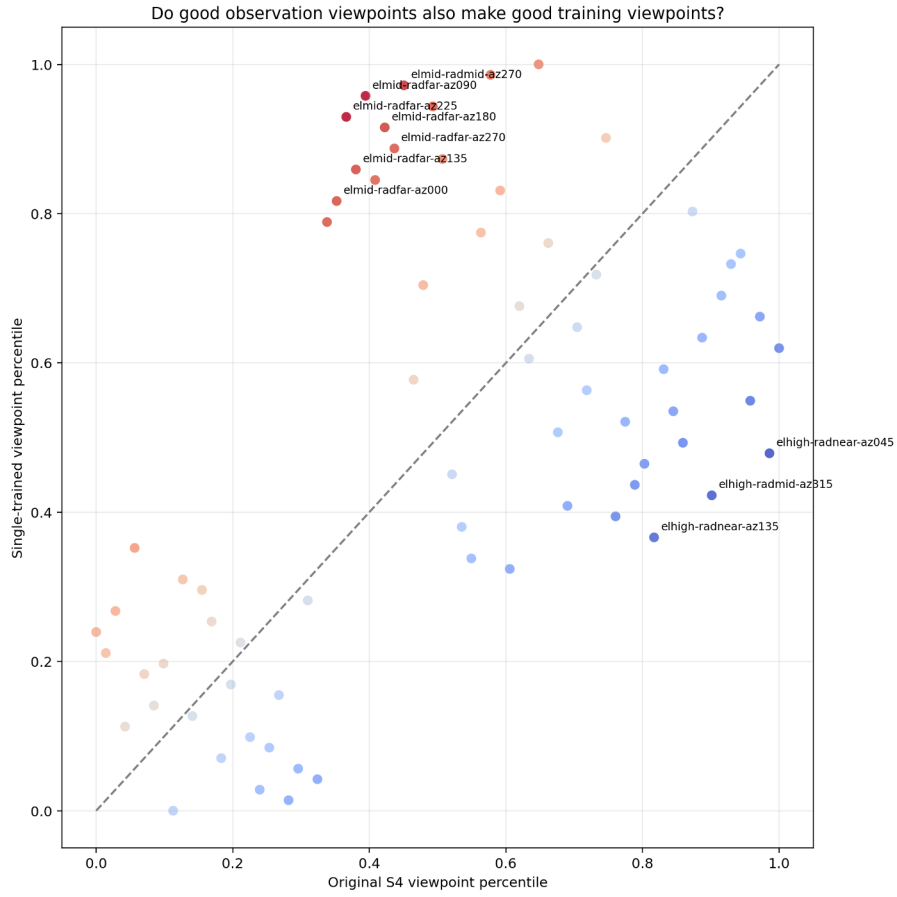


Figure A.6: Observation-view strength versus training-view strength across the 72 viewpoints. Points away from the diagonal indicate that viewpoints can differ in their usefulness as test-time observations and as single-view training sources.

A.3.2 Experiment 2: Fixed-Detector Viewpoint Sensitivity

Exact Azimuth–Elevation Heatmaps by Class

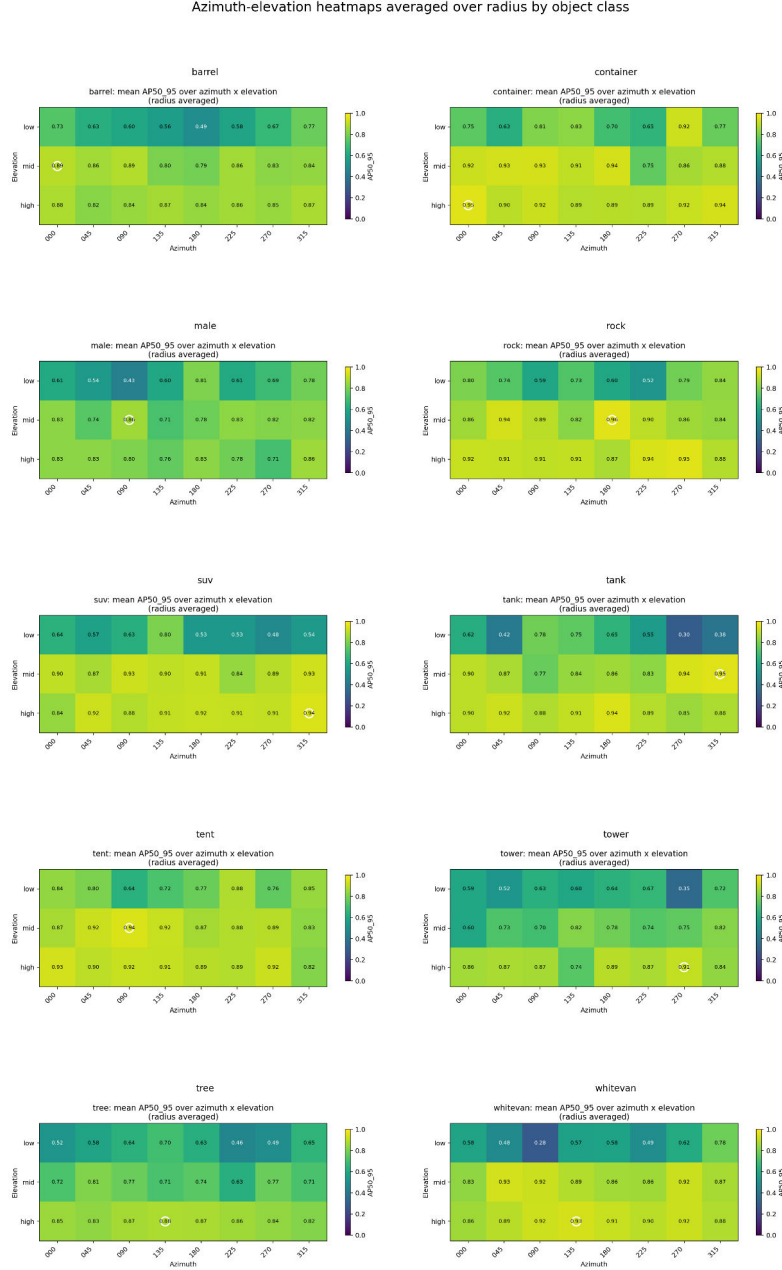


Figure A.7: Exact azimuth–elevation heatmaps averaged over radius for all object classes. Each panel exposes the detailed surface for one class after averaging over radius, which is useful for spotting local pockets and class- specific asymmetries. These plots are informative exploratory material, but they are less statistically stable than the factor-level aggregation used in the main text.

A.3.3 Experiment 3: Inference-Time Multiview Analysis

Multiview Threshold-Protocol Detail

Table A.9 and Figure A.8 report the full 1-of- k threshold-protocol results underlying the summary in Section 4; these values are also included in harmonized form in the coalition-method comparison below (Table A.10).

Table A.9: Operational threshold-protocol results with the detector held fixed.

Setting	Binary target found	Target AP _{50:95}	Target strict quality
1-of-1	0.9811 ± 0.0037	0.8516 ± 0.0077	0.8732 ± 0.0077
1-of-2 OR	0.9986 ± 0.0005	0.9277 ± 0.0058	0.9088 ± 0.0055
2-of-2 confirmation	0.9635 ± 0.0071	0.7755 ± 0.0103	0.8375 ± 0.0103
1-of-3 OR	0.9999 ± 0.0001	0.9487 ± 0.0050	0.9189 ± 0.0050
2-of-3 confirmation	0.9961 ± 0.0015	0.8858 ± 0.0077	0.8887 ± 0.0068
3-of-3 unanimous	0.9473 ± 0.0101	0.7204 ± 0.0121	0.8119 ± 0.0125



Figure A.8: One-view versus two-view operational comparison under the stricter target-quality scoring used in the repository.

Full Coalition-Method Comparison

Access to additional observations strongly improves availability-oriented measures under the least restrictive (1-of- k OR) threshold rule, while stricter confirmation rules that require multiple views to agree perform worse on these same metrics, reflecting a different, more conservative decision criterion rather than a failure of multiview reasoning. Table A.10 compares selection-based and accumulation-based fusion on a shared scene-balanced strict-quality scale.

Table A.10: Harmonized comparison of coalition methods, organized by the number of available views. AP50 : 95 and found rate are only methodologically comparable within the threshold family; dashes indicate that the metric is not used for that method family.

Family	Method	Strict quality	Δ vs. 1-view	AP _{50:95}	Found rate
1 view					
Reference	Single-view reference	0.8732	+0.0000	0.8516	0.9811
2 views					
Noisy-OR fusion	Noisy-OR + best IoU	0.9515	+0.0784	–	–
Odds-product fusion	Odds-product + best IoU	0.9487	+0.0756	–	–
Noisy-OR fusion	Noisy-OR + mean IoU	0.9356	+0.0625	–	–
Odds-product fusion	Odds-product + mean IoU	0.9330	+0.0598	–	–
Conservative corroboration	Support-weighted noisy-OR	0.9212	+0.0481	–	–
Threshold	1-of-2 OR	0.9088	+0.0357	0.9277	0.9986
Selection / rescue	Best box (max)	0.9088	+0.0357	–	–
Strict confirmation	Unanimous best box	0.8801	+0.0069	–	–
Naive aggregation	Mean quality	0.8732	+0.0000	–	–
Threshold	2-of-2 confirmation	0.8375	-0.0357	0.7755	0.9635
3 views					
Noisy-OR fusion	Noisy-OR + best IoU	0.9661	+0.0929	–	–
Odds-product fusion	Odds-product + best IoU	0.9643	+0.0912	–	–
Noisy-OR fusion	Noisy-OR + mean IoU	0.9436	+0.0705	–	–
Odds-product fusion	Odds-product + mean IoU	0.9420	+0.0689	–	–
Conservative corroboration	Support-weighted noisy-OR	0.9269	+0.0538	–	–
Threshold	1-of-3 OR	0.9189	+0.0457	0.9487	0.9999
Selection / rescue	Best box (max)	0.9189	+0.0457	–	–
Threshold	2-of-3 confirmation	0.8887	+0.0156	0.8858	0.9961
Strict confirmation	Unanimous best box	0.8743	+0.0012	–	–
Naive aggregation	Mean quality	0.8732	+0.0000	–	–
Threshold	3-of-3 unanimous	0.8119	-0.0613	0.7204	0.9473

Challenge-Slice Analysis

Figure A.9 tests where evidence accumulation matters most by comparing top fusion methods against best-box selection on harder slices — weak primary detections, small target boxes, low primary IoU, far primary views, and hard scenes — referenced in Section 4.

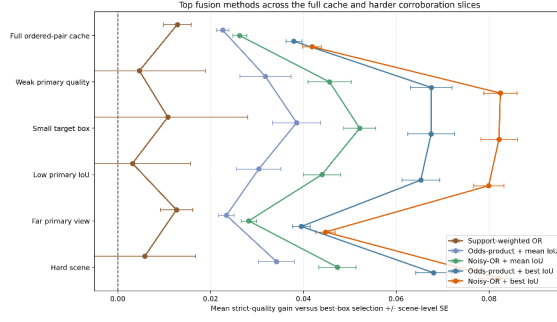


Figure A.9: Fusion gains under harder target conditions. The figure shows the mean strict-quality gain of the top accumulation-based fusion methods relative to best-box selection, across the full ordered-pair cache and five harder challenge slices. Positive values indicate that the fusion rule improves target strict quality beyond selecting the best matched box.

A.3.4 Experiment 4: Real-World Transfer and Adaptation

Full Real-World Transfer Results

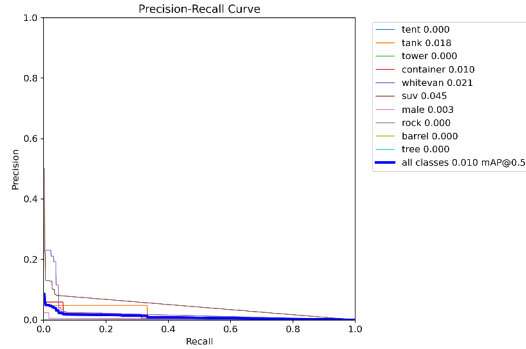
Table A.11 reports the complete per-class breakdown across all four zero-shot transfer settings referenced in Section 4; the aggregate ("all") row for each dataset is reproduced in the main text.

Table A.11: Combined per-class overview of the four real-world transfer experiments. VisDrone and AU-AIR are approximate overlapping-class transfer tests in which real classes were mapped to the synthetic taxonomy. The same-taxonomy real-image dataset uses the ten synthetic classes but remains small and exploratory. The self-collected UAV validation set adds a physical drone-capture test using the same broad object taxonomy.

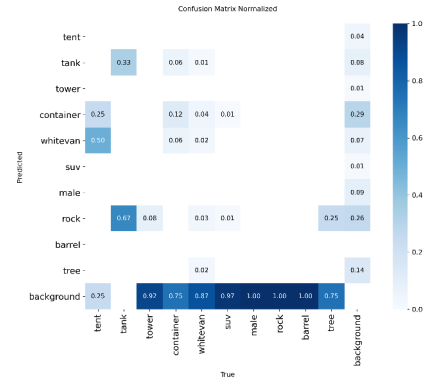
Class	Images	Instances	Precision	Recall	mAP50	mAP50 : 95
VisDrone						
<i>all</i>	277	9,525	0.0479	0.0114	0.0111	0.0039
<i>whitevan</i>	277	1,381	0.0000	0.0000	0.0000	0.0000
<i>suv</i>	277	5,507	0.0376	0.0013	0.0153	0.0075
<i>male</i>	277	2,637	0.1060	0.0330	0.0178	0.0043
AU-AIR						
<i>all</i>	32,823	127,265	0.0281	0.0136	0.0038	0.0011
<i>whitevan</i>	13,634	19,532	0.0055	0.0005	0.0003	0.0001
<i>suv</i>	29,381	102,581	0.0389	0.0022	0.0047	0.0018
<i>male</i>	4,709	5,152	0.0398	0.0380	0.0064	0.0015
Same Object Dataset						
<i>all</i>	54	235	0.434	0.0555	0.0502	0.0269
<i>tent</i>	9	33	0.000	0.0000	0.0103	0.0027
<i>tank</i>	4	6	1.000	0.0000	0.0054	0.00486
<i>tower</i>	4	4	1.000	0.0000	0.0000	0.0000
<i>container</i>	4	19	0.137	0.1580	0.0395	0.0158
<i>whitevan</i>	7	8	0.000	0.0000	0.00181	0.000362
<i>suv</i>	13	28	0.247	0.0357	0.0785	0.0276
<i>male</i>	8	42	0.557	0.0714	0.119	0.0174
<i>rock</i>	4	8	0.248	0.25	0.218	0.186
<i>barrel</i>	5	55	1.000	0.0000	0.0000	0.0000
<i>tree</i>	4	32	0.152	0.398	0.0299	0.0141
Self-collected UAV validation						
<i>all</i>	156	592	0.1440	0.0501	0.0098	0.0044
<i>tent</i>	4	4	0.0000	0.0000	0.0000	0.0000
<i>tank</i>	3	3	0.0320	0.3330	0.0181	0.0030
<i>tower</i>	12	12	0.0000	0.0000	0.0000	0.0000
<i>container</i>	6	16	0.0203	0.1250	0.0101	0.0050
<i>whitevan</i>	64	126	0.1780	0.0397	0.0212	0.0108
<i>suv</i>	80	344	0.2070	0.0029	0.0454	0.0235
<i>male</i>	30	62	0.0000	0.0000	0.0028	0.0017
<i>rock</i>	9	9	0.0000	0.0000	0.0000	0.0000
<i>barrel</i>	4	8	1.0000	0.0000	0.0000	0.0000
<i>tree</i>	4	8	0.0000	0.0000	0.0000	0.0000

Self-Collected UAV Diagnostic Plots

Figure A.10 shows the precision–recall curve and normalized confusion matrix for the self-collected UAV zero-shot evaluation reported in Section 4.



(a) Precision-recall curve.



(b) Normalized confusion matrix.

Figure A.10: Diagnostic plots for the self-collected UAV validation run. The all-class mAP50 is approximately 0.010, and the confusion matrix shows that missed detections dominate the error profile.

Real Same-Object Exact Shapley: Single-View Quality vs. Contribution

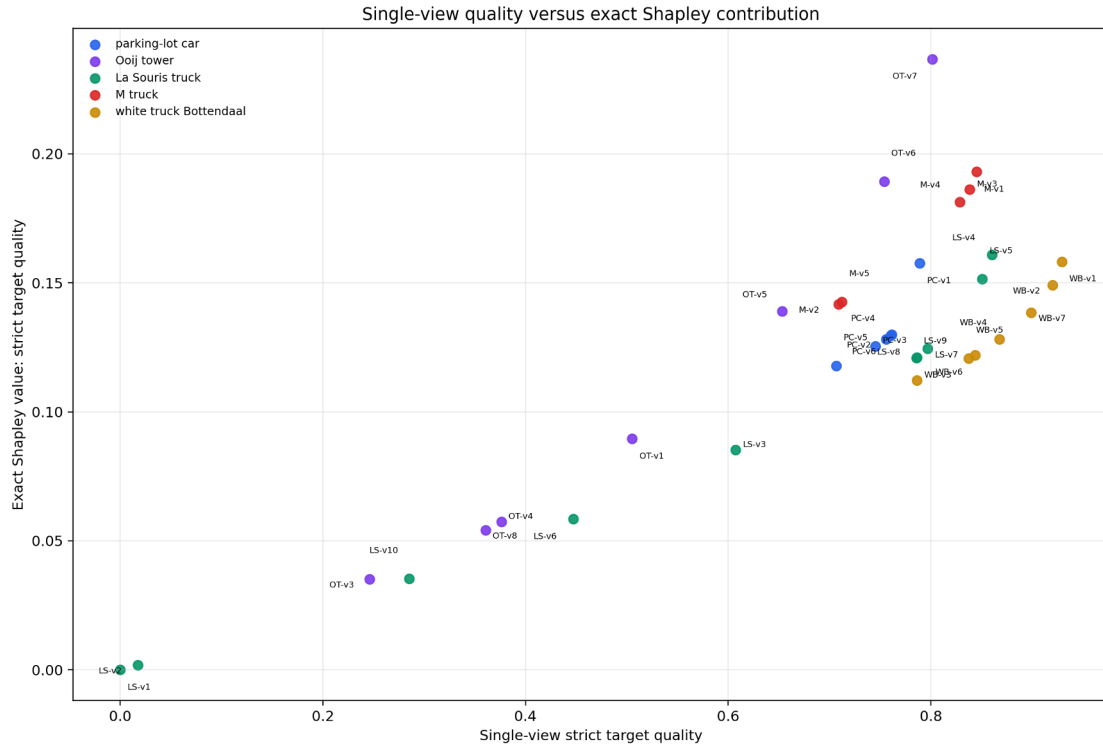


Figure A.11: Relationship between single-view strict target quality and exact strict-quality Shapley contribution for the fine-tuned detector. The strong descriptive association reflects the maximum-quality coalition rule, under which views that are strong individually tend to improve a larger number of coalitions.

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